Advanced Computer-Aided VLSI System Design

APB and AHB

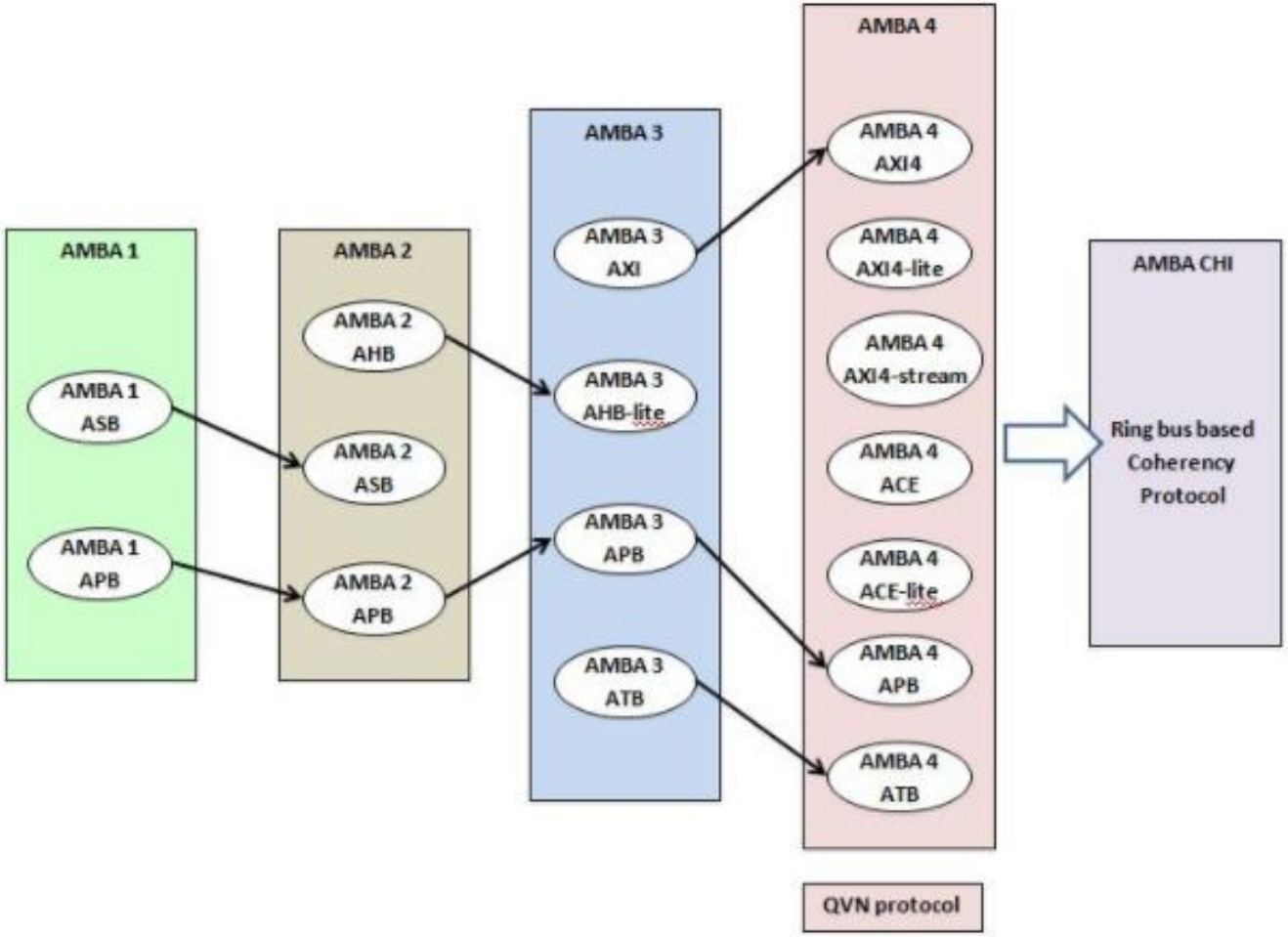
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Graduate Institute of Electronics Engineering, National Taiwan University



NTU GIEE

Evolution of the AMBA Protocol



Advanced Microcontroller Bus Architecture (AMBA)



Key AMBA Protocols

- AMBA Advanced Peripheral Bus (AMBA APB)
  - APB is highly compact, low power, and allows configuration and low-bandwidth traffic to be isolated from high-performance interconnects
- AMBA Advanced High Performance Bus (AMBA AHB)
  - AHB defines an interface protocol that is widely used for embedded designs and low-latency SoCs
- AMBA Advanced Extensible Interface (AMBA AXI)
  - AXI defines protocols for implementing high-frequency, high-bandwidth, and low-latency interconnect designs



Outline

- APB (Advanced Peripheral Bus)
- AHB (Advanced High-performance Bus)
  - Managers and Subordinates
  - Interconnect
  - Signals
  - Transfer
  - Burst
  - Multi-layer AHB



APB (Advanced Peripheral Bus)

- Primarily used in low-bandwidth peripheral devices, such as UART, I2C, and PIO (Programmed I/O)
- Single master: The master is typically an APB bridge that connects AXI/AHB to APB
- It requires two cycles for read/write operations (If the slave is ready)
- It does not support pipelined transfer
- Versions: APB5, APB4, APB3, APB2

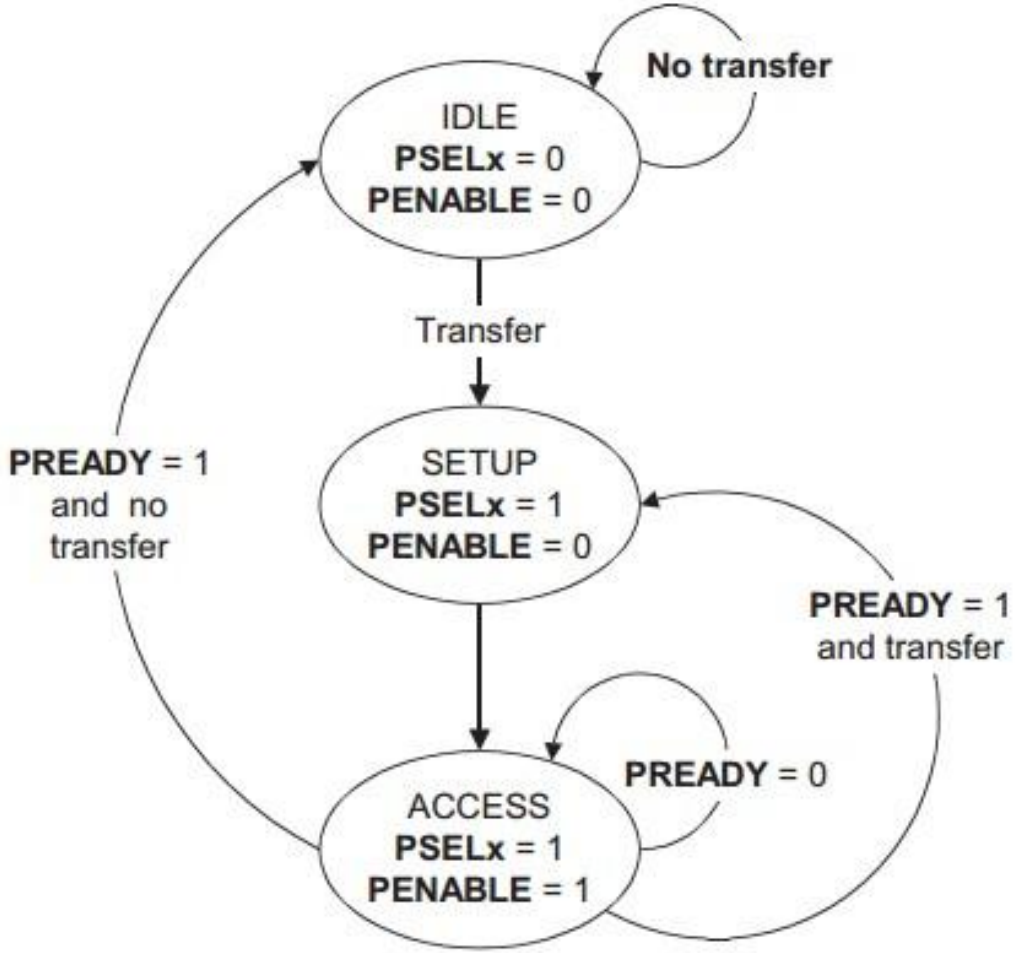


APB Signals

Signal	Width	Source	APB5	APB4	APB3	APB2	Description
PCLK	1	Clock	Y	Y	Y	Y	Clock
PRESETn	1	Reset	Y	Y	Y	Y	Negative reset
PADDR	ADDR_WH	M	Y	Y	Y	Y	Address (maximal 32 bits)
PSELx	1	M	Y	Y	Y	Y	Select
PENABLE	1	M	Y	Y	Y	Y	Enable
PWRITE	1	M	Y	Y	Y	Y	1: Write, 0: Read
PWDATA	DATA_WH	M	Y	Y	Y	Y	Write data (8, 16, 32 bits)
PSTRB	DATA_WH/8	M	Y	Y	N	N	Write strobe PSTRB[n] -> PWDATA[(8n + 7):(8n)] Must be 0 during a read transfer
PREADY	1	S	Y	Y	Y	N	Ready
PRDATA	DATA_WH	S	Y	Y	Y	Y	Read data (8, 16, 32 bits)
PSLVERR	1	S	Y	Y	Y	N	Transfer error
PWAKEUP	1	M	Y	N	N	N	Can be routed to a clock controller, or similar component, to enable power and clocks to connected components

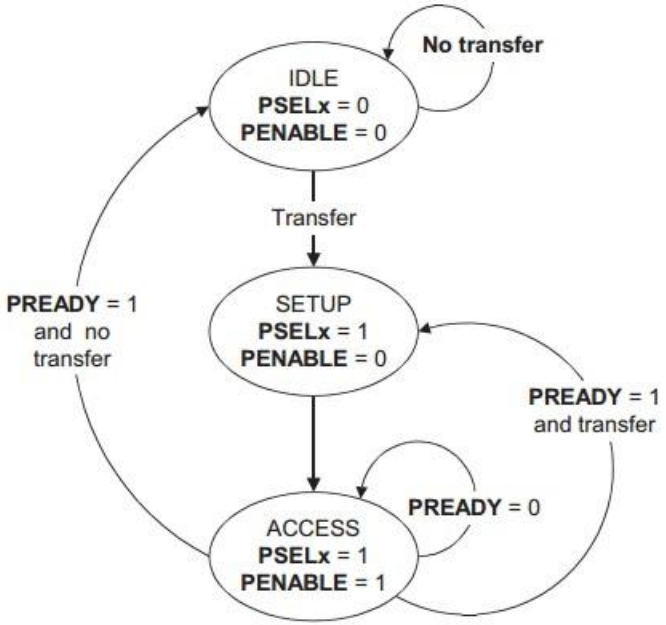
* The protection signals for APB5 and APB4, as well as the user-defined and parity check signals for APB5, are not listed.

Operating States



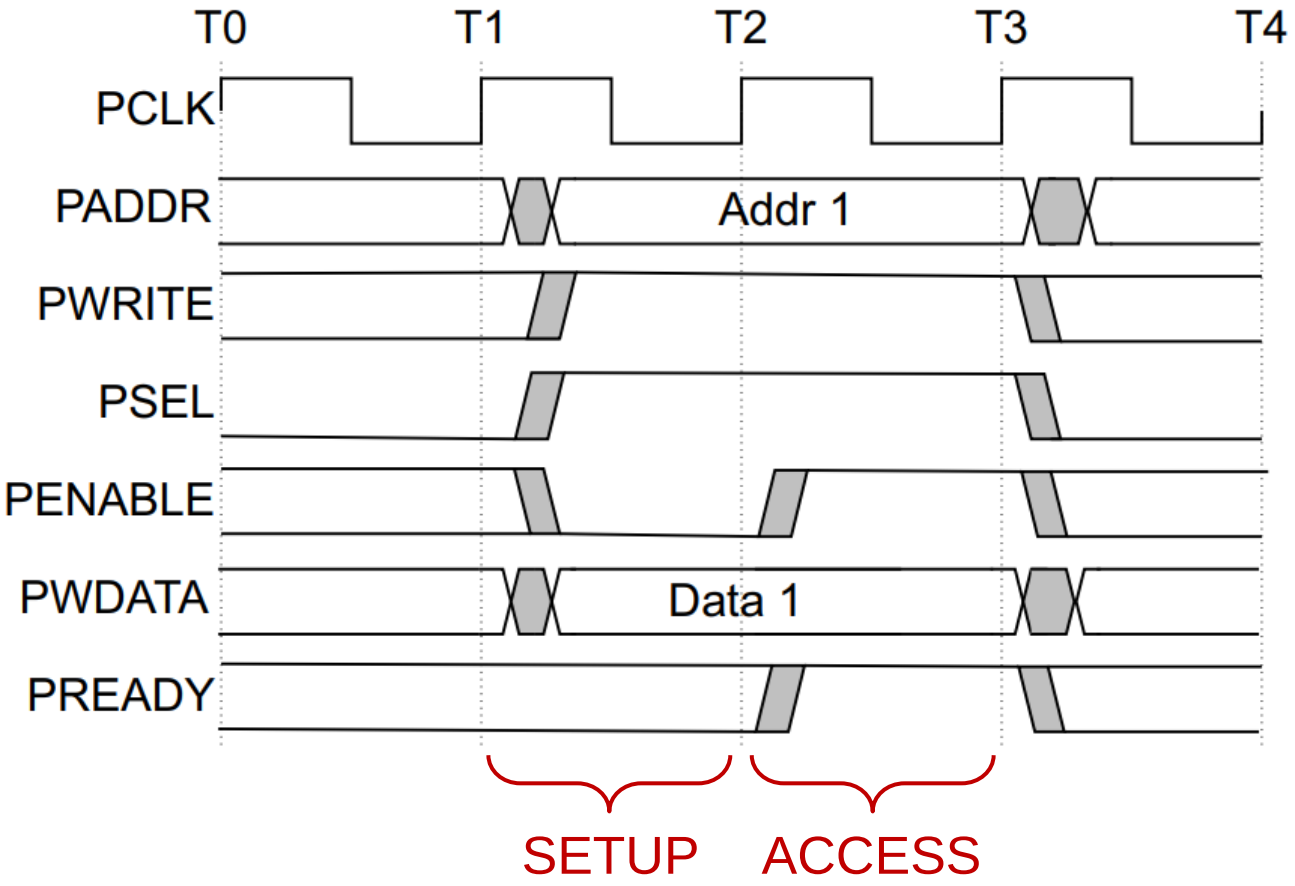
Operating States

- IDLE: default state
- SETUP
  - The interface moves into the SETUP state when a transfer is required
  - One clock cycle
- ACCESS
  - The exit from the ACCESS state is controlled by PREADY from the peripheral
  - PREADY = 1: exit
  - PREADY = 0: stay



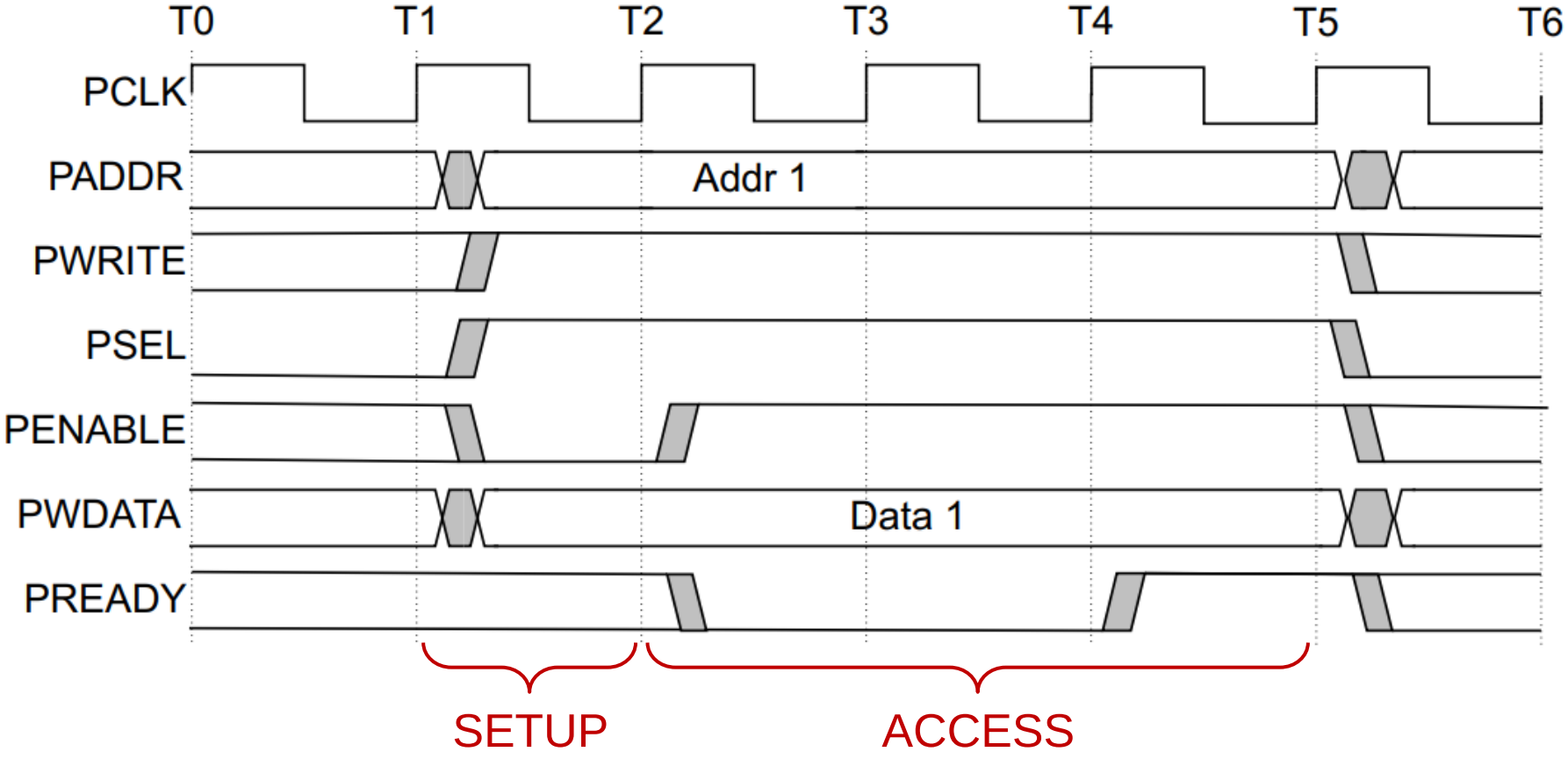


Write: No Wait





Write: Wait

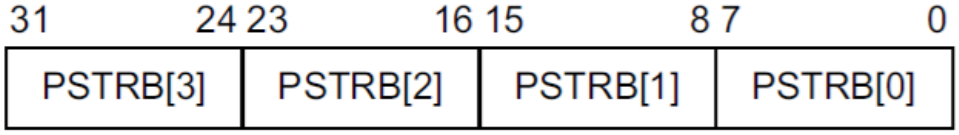


- The peripheral extends the transfer by driving PREADY low



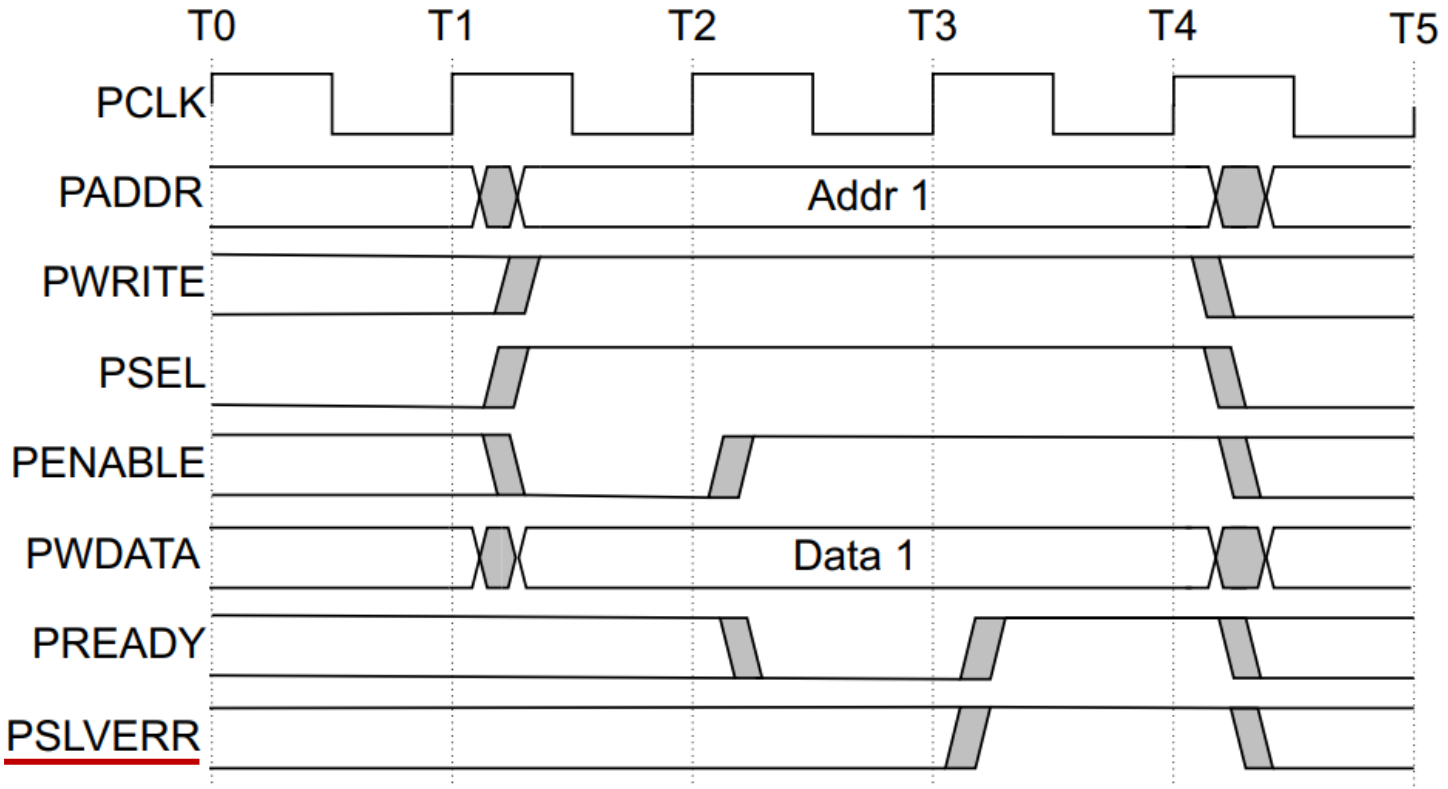
Write: Strobes

- PSTRB enables sparse data transfer on the write data bus
- PSTRB indicates that the corresponding byte lane of the write data bus contains valid information



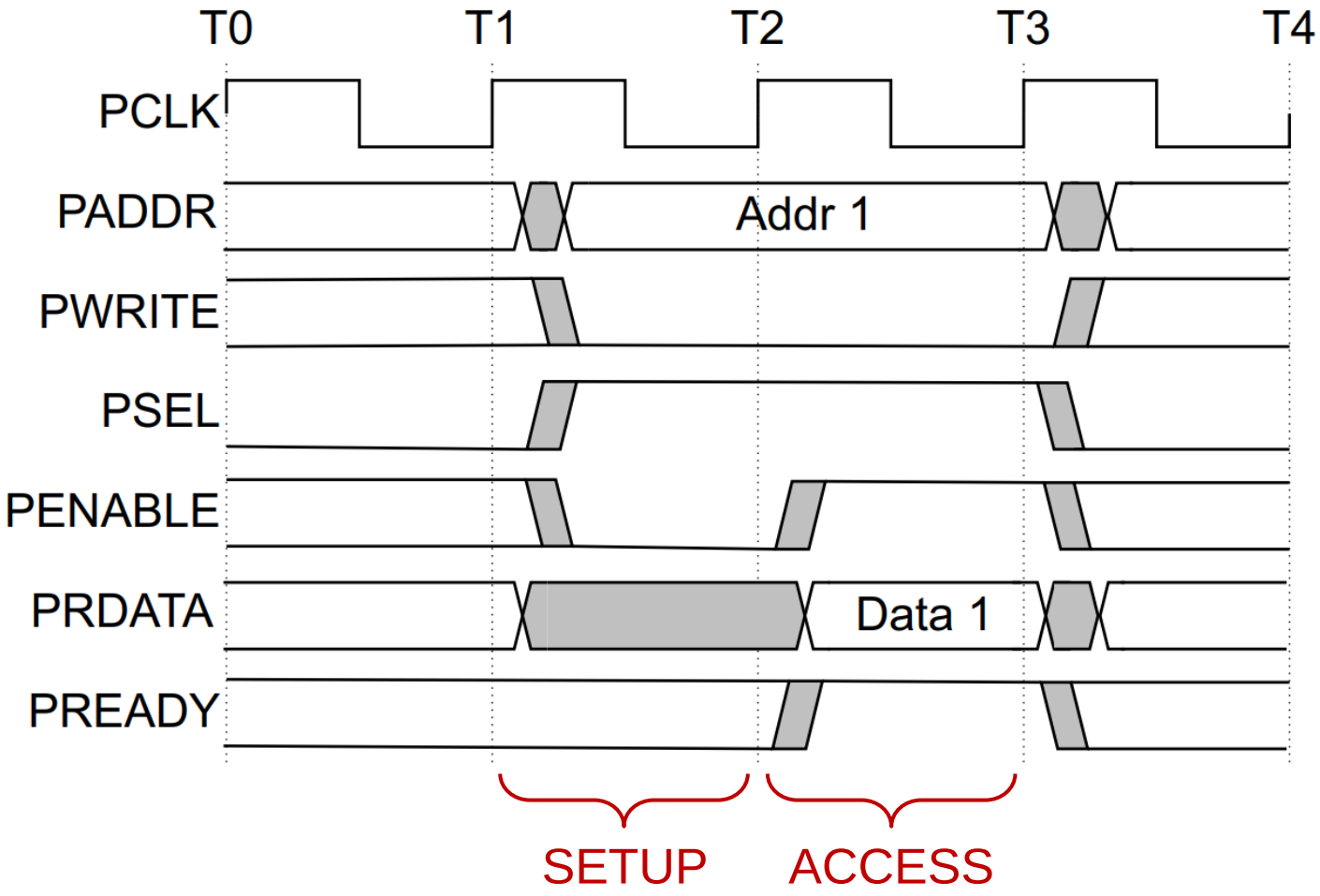
- For read transfers, the master must drive all bits of PSTRB low

Write: Error


T0
T1
T2
T3
T4
T5
PCLK
PADDR
Addr 1
PWRITE
PSEL
PENABLE
PWDATA
Data 1
PREADY
PSLVERR

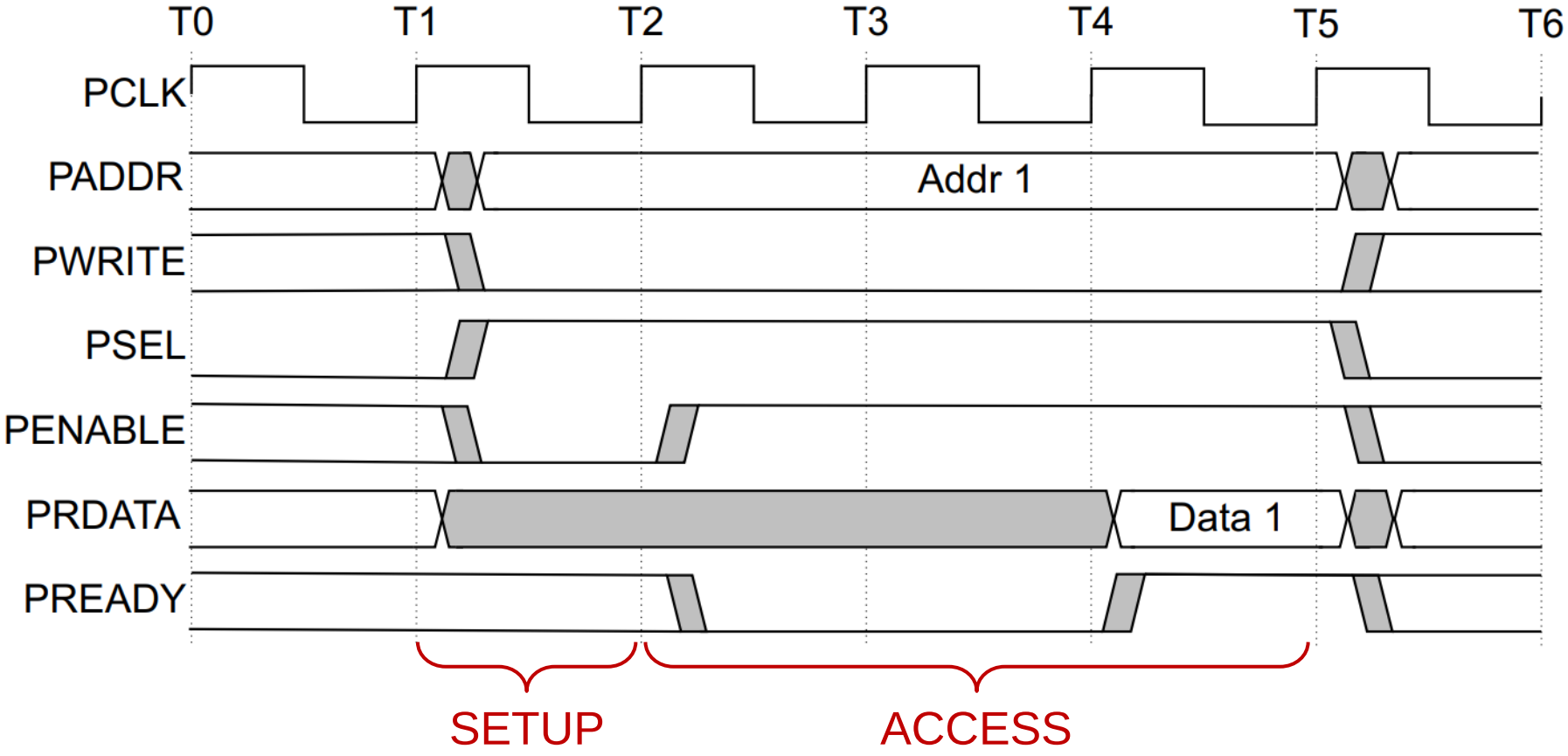


Read: No Wait


T0
T1
T2
T3
T4
PCLK
PADDR
Addr 1
PWRITE
PSEL
PENABLE
PRDATA
Data 1
PREADY
SETUP
ACCESS



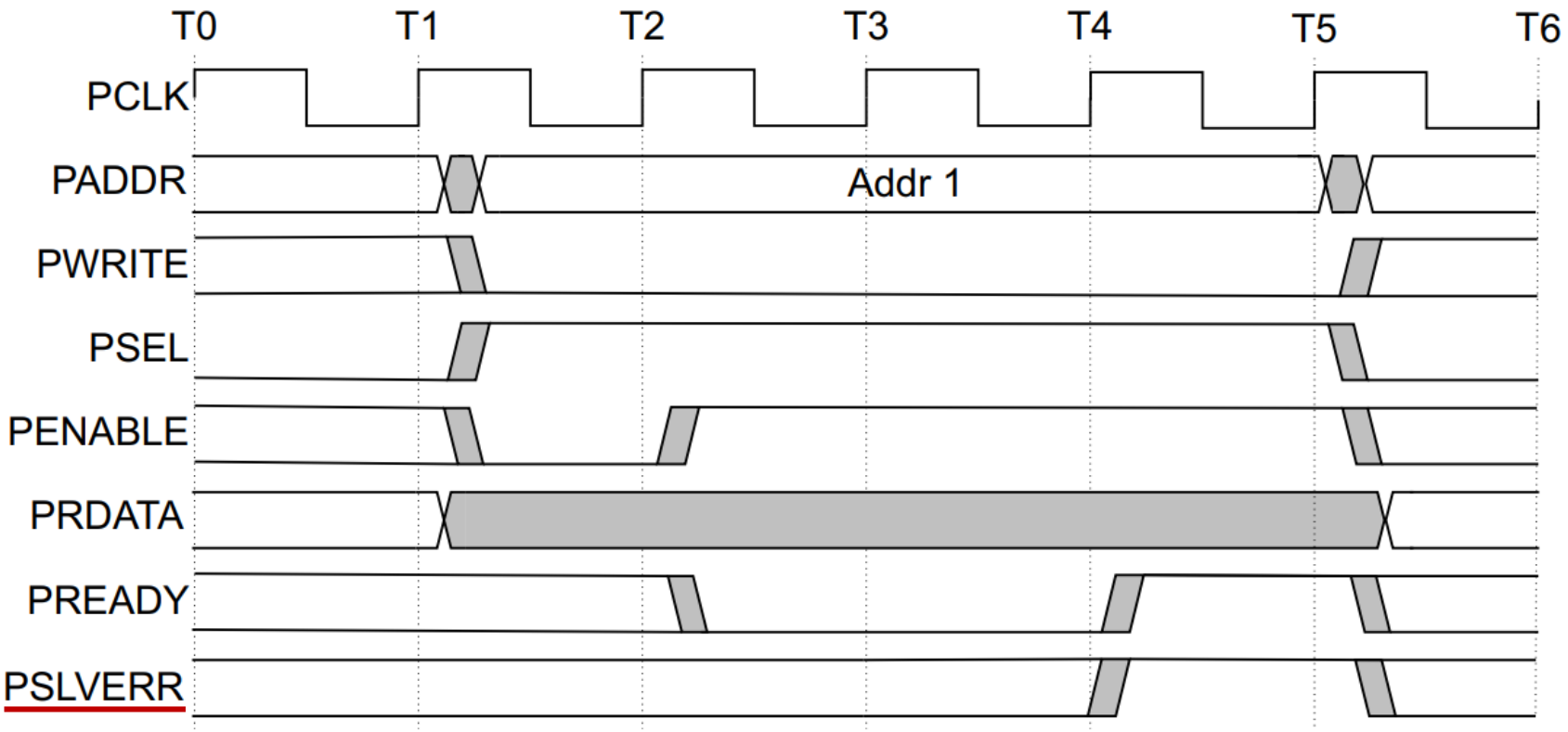
Read: Wait



- The peripheral extends the transfer by driving PREADY low



Read: Error


T0
T1
T2
T3
T4
T5
T6
PCLK
PADDR
Addr 1
PWRITE
PSEL
PENABLE
PRDATA
PREADY
PSLVERR



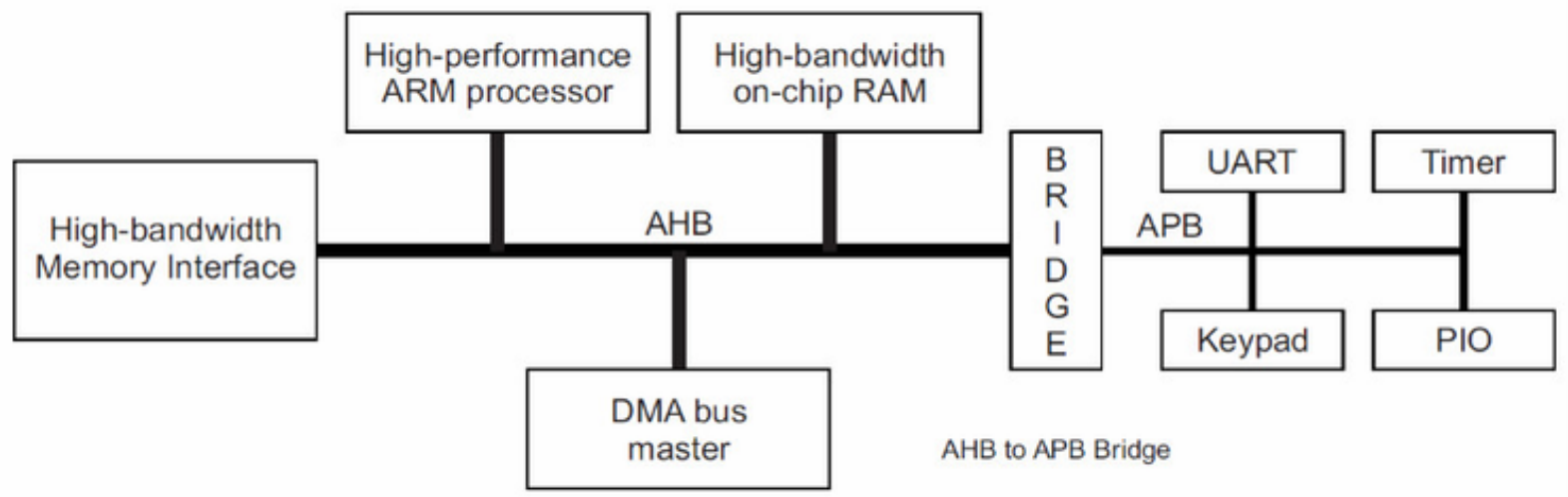
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AHB (Advanced High-performance Bus)

- Primarily utilized in high-efficiency and high-clock-frequency system architectures, including CPUs, RAM, NAND flash, DMA, bridges, etc

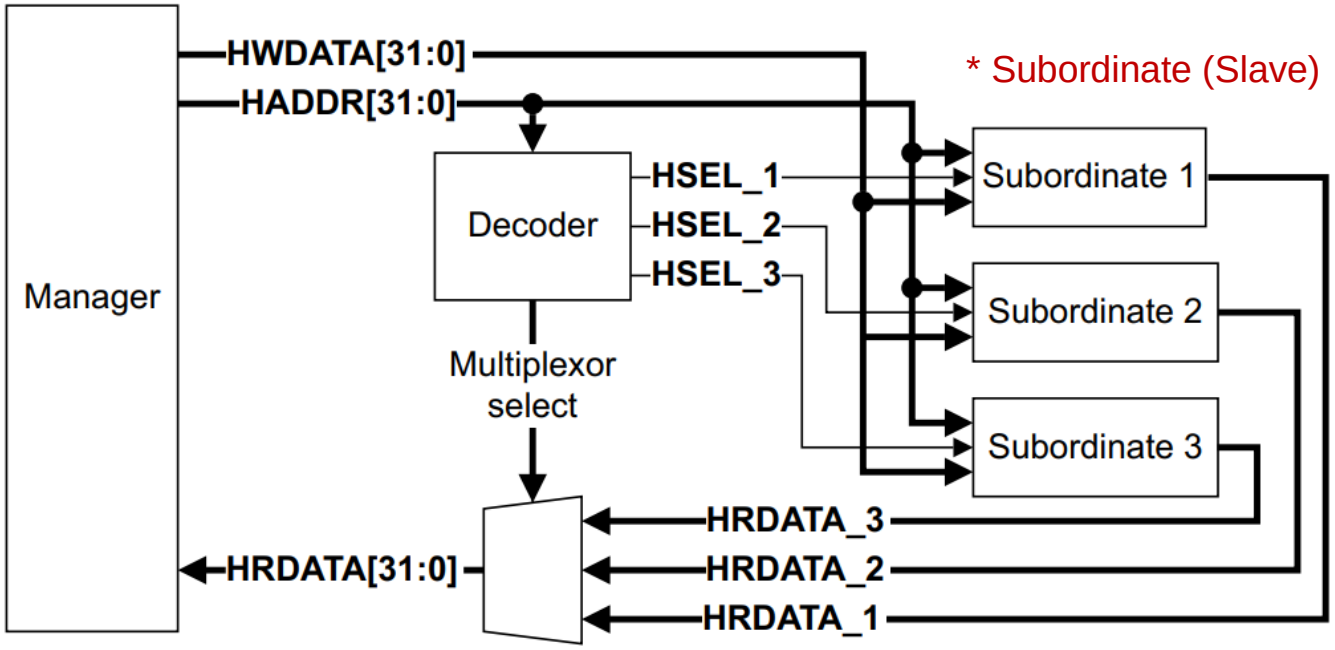


AHB Block Diagram

- AHB manager: The main component of AHB
- Decoder: Selects the subordinate
- Multiplexer: Routes the subordinate's output data back

* Manager (Master)

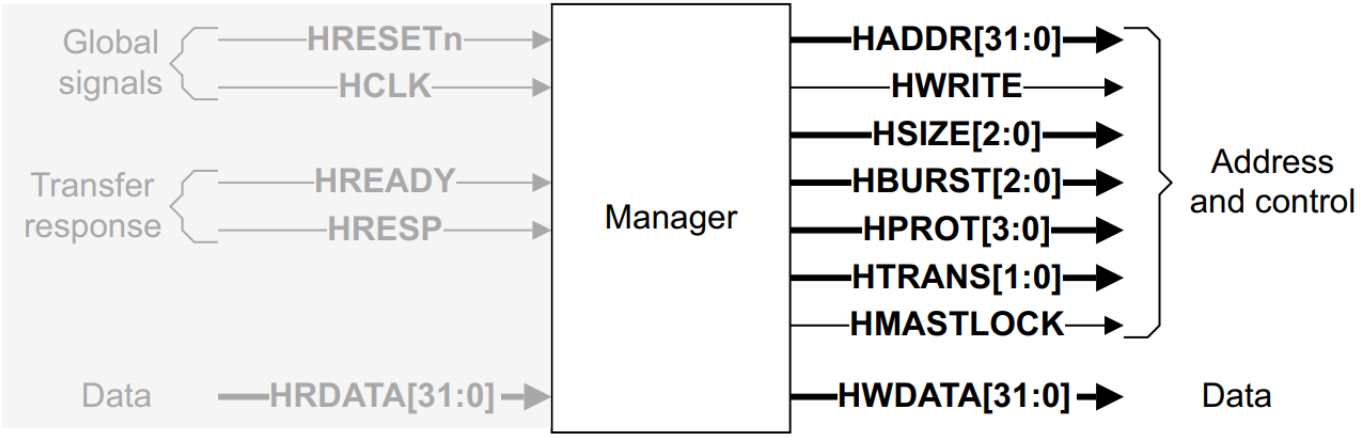
* 1 master and 3 slaves without matrix interconnect





Manager Interface: Output

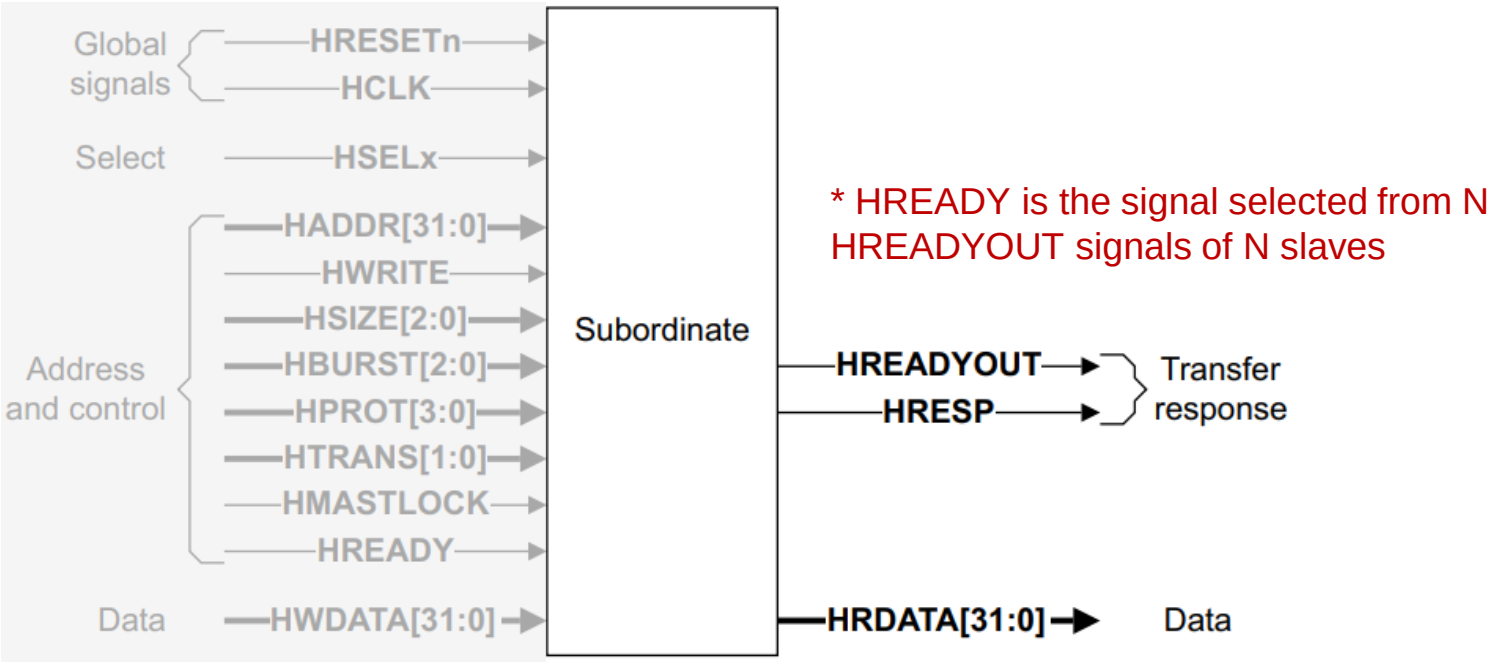
Signal	Width	Destination	Description
HADDR	ADDR_WH	Slave and Decoder	10~64 bits, commonly fixed at 32 bits
HWRITE	1	Slave	1: Write, 0: Read
HSIZE	3	Slave	Size of the transfer
HBURST	HBURST_WH	Slave	Burst type
HPROT	HPROT_WH	Slave	Protection control signal (See documents)
HTRANS	2	Slave	Transfer type: IDLE, BUSY, NONSEQ, SEQ
HMASTLOCK	1	Slave	Indication of locked sequence (See documents)
HWDATA	DATA_WH	Slave	2^N (N: 3~10) width, 32~256 is recommended





Subordinate Interface: Output

Signal	Width	Destination	Description
HREADYOUT	1	Multiplexer	1: transfer finishes, 0: extends the transfer
HRESP	1	Multiplexer	0: OKAY status, 1: ERROR status
HRDATA	DATA_WH	Multiplexer	2^N (N: 3~10) width, 32~256 is recommended





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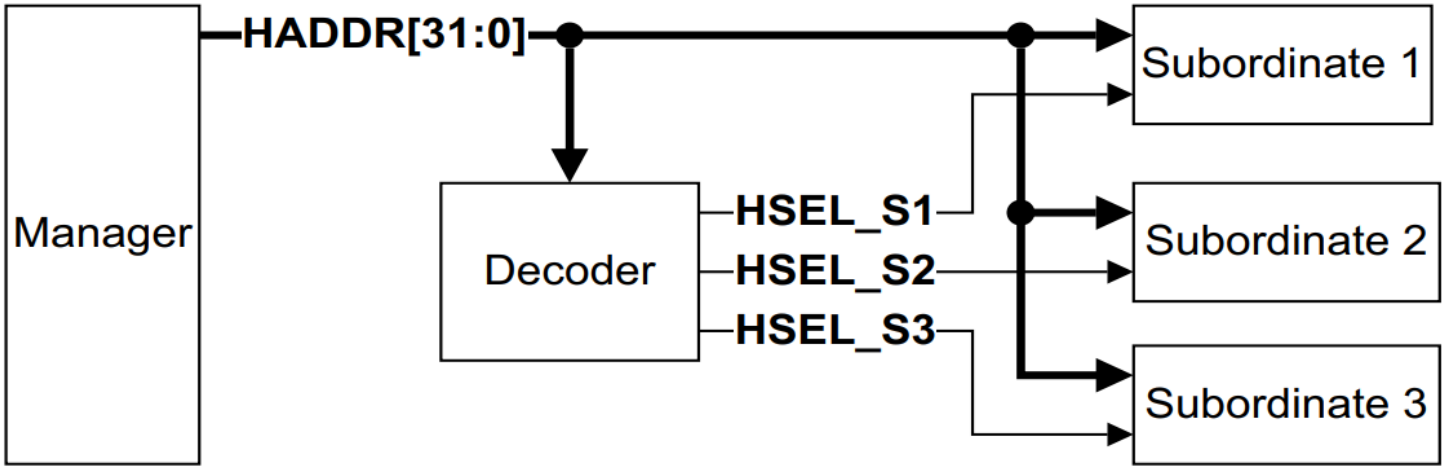
Interconnect

- An interconnect component provides the connection between managers and subordinates in a system
- A single-manager system only requires the use of a decoder and a multiplexer
- A multi-manager system requires an interconnect that provides arbitration and routes signals from different managers to the appropriate subordinates

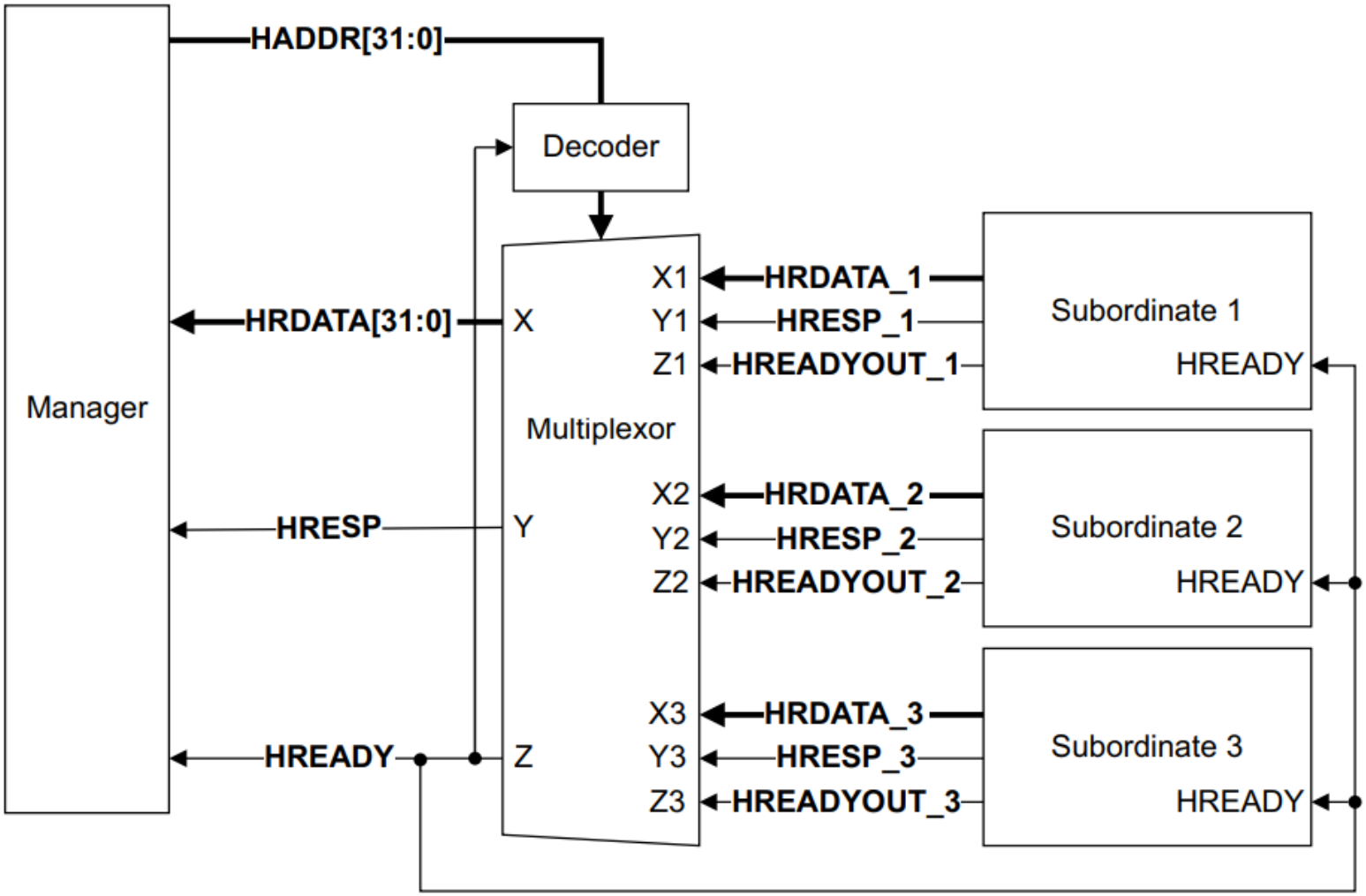


Decoder

- Decodes the address of each transfer and provides a select signal for the subordinate
- Provides a control signal to the multiplexer
- A single centralized decoder is required in all implementations that use two or more subordinates



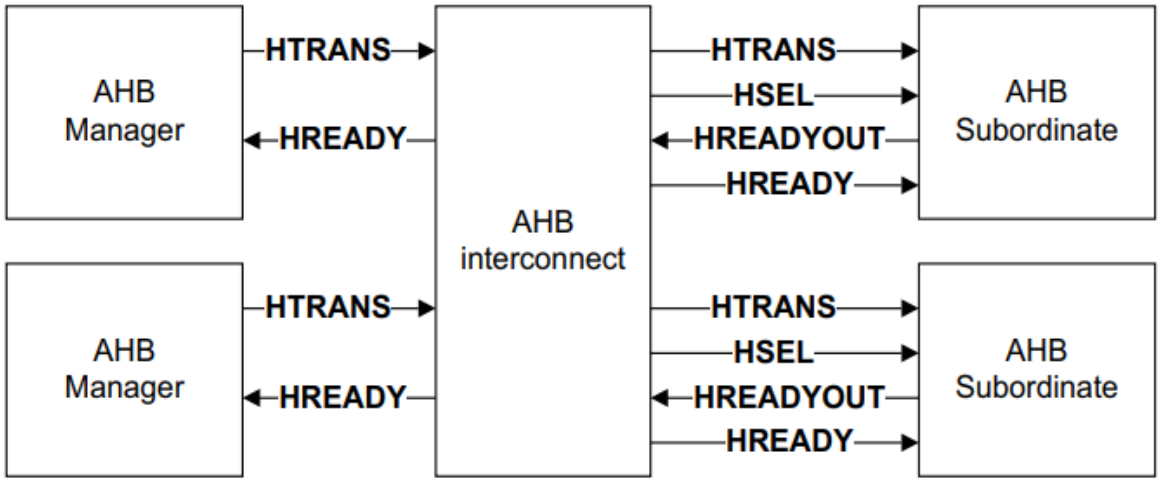
Multiplexer





Interconnect with AHB Interfaces

- A generic interconnect has HTRANS, HREADY, and HSEL



- The HREADY output from the interconnect can be used to stall a manager or a subordinate
  - A subordinate has inserted wait states
  - A manager is waiting for arbitration
  - Previous transfer is stalled



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HSIZE

- HSIZE[2:0] indicates the size of a data transfer

HSIZE[2]	HSIZE[1]	HSIZE[0]	Size (bits)	Description
0	0	0	8	Byte
0	0	1	16	Halfword
0	1	0	32	Word
0	1	1	64	Doubleword
1	0	0	128	4-word line
1	0	1	256	8-word line
1	1	0	512	-
1	1	1	1024	-

- The transfer size set by HSIZE must <= the data bus width
  - With a 32-bit data bus, HSIZE must only use the values 0b000, 0b001, or 0b010



HBURST

- HBURST[2:0] indicates how many transfers are in the burst and how the address increments

HBURST[2:0]	Type	Description
0b000	SINGLE	Single transfer burst
0b001	INCR	Incrementing burst of undefined length
0b010	WRAP4	4-beat wrapping burst
0b011	INCR4	4-beat incrementing burst
0b100	WRAP8	8-beat wrapping burst
0b101	INCR8	8-beat incrementing burst
0b110	WRAP16	16-beat wrapping burst
0b111	INCR16	16-beat incrementing burst

- Managers must not attempt to start an incrementing burst that crosses a 1KB address boundary



Burst Termination After a Busy Transfer

- During a burst, the manager uses BUSY transfers if it needs more time before the next transfer in the burst
- For an undefined length burst, INCR, the manager can insert BUSY transfers, and then use a NONSEQ or IDLE transfer to terminate the burst
- The manager is not permitted to end a fixed-length burst (INCRX, WRAPX) with a BUSY transfer
- The manager is not permitted to perform a BUSY transfer immediately after a SINGLE burst
  - Must be followed by an IDLE or a NONSEQ transfer



Early Burst Termination by the Subordinate

- If a subordinate provides an ERROR response, then the manager can cancel the remaining transfers in the burst
- If the manager cancels the remaining transfers in the burst, then it must change HTRANS to indicate IDLE during the two-cycle ERROR response
- If the manager does not complete the burst, there is no requirement for it to rebuild the burst when it next accesses that subordinate
  - If a manager only completes 3 beats of an 8-beat burst, it does not have to complete the remaining 5 transfers



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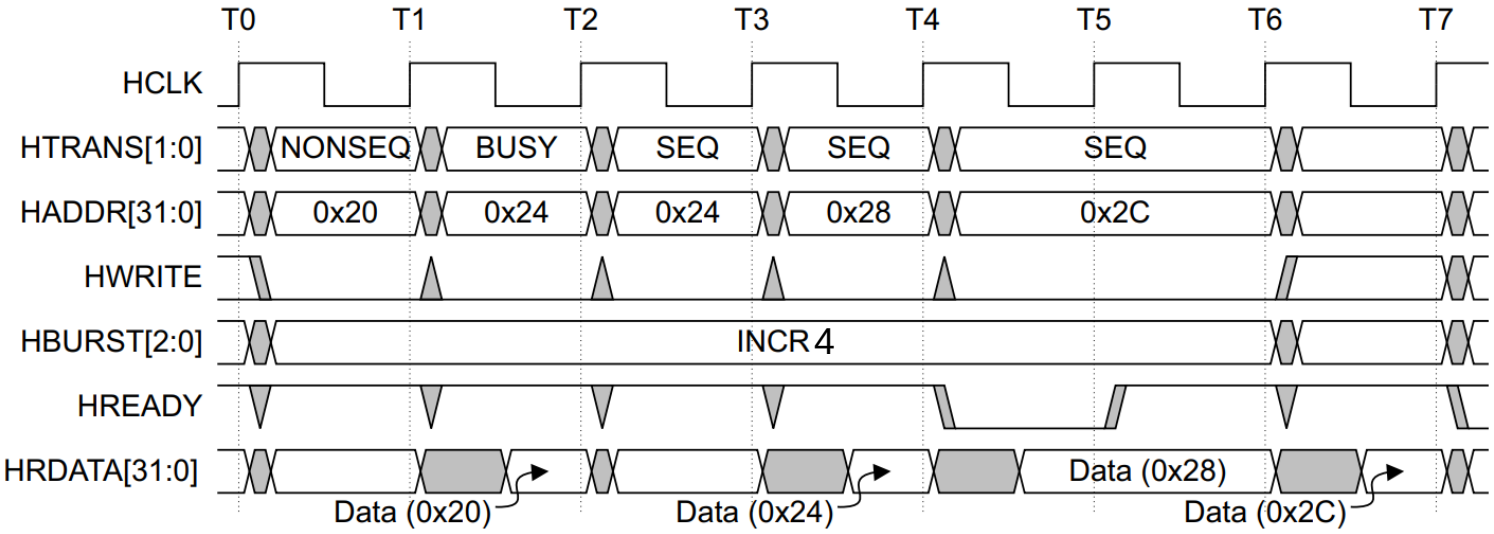


HTRANS

HTRANS[1:0]	Type	Description
00	IDLE	Indicates that no data transfer is required
01	BUSY	The BUSY transfer type enables managers to insert idle cycles in the middle of a burst
10	NONSEQ	Indicates a single transfer or the first transfer of a burst
11	SEQ	The remaining transfers in a burst are SEQUENTIAL and the address is related to the previous transfer



Example: HTRANS (1/2)



T0-T1 The 4-beat read starts with a NONSEQ transfer.

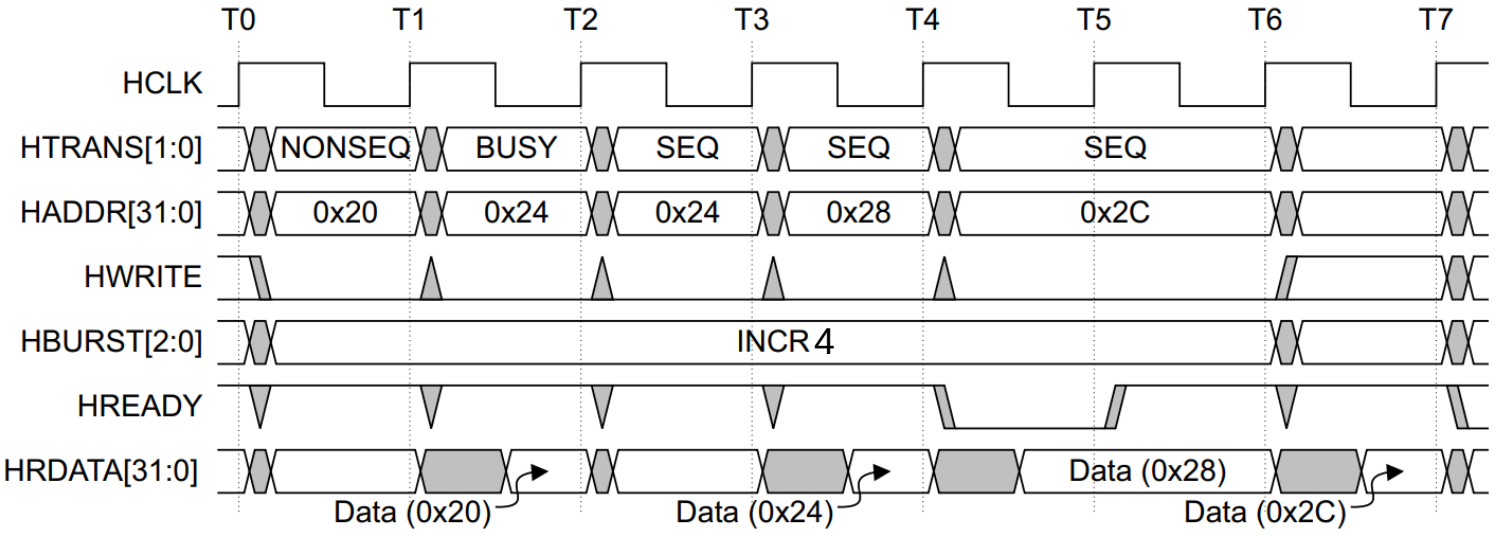
T1-T2 The Manager is unable to perform the second beat and inserts a BUSY transfer to delay the start of the second beat.

The Subordinate provides the read data for the first beat.

T2-T3 The Manager is now ready to start the second beat, so a SEQ transfer is signaled. The Manager ignores any data that the Subordinate provides on the read data bus.



Example: HTRANS (2/2)



T3-T4 The Manager performs the third beat.

The Subordinate provides the read data for the second beat.

T4-T5 The Manager performs the last beat.

The Subordinate is unable to complete the transfer and uses **HREADYOUT** to insert a single wait state.

T5-T6 The Subordinate provides the read data for the third beat.

T6-T7 The Subordinate provides the read data for the last beat.



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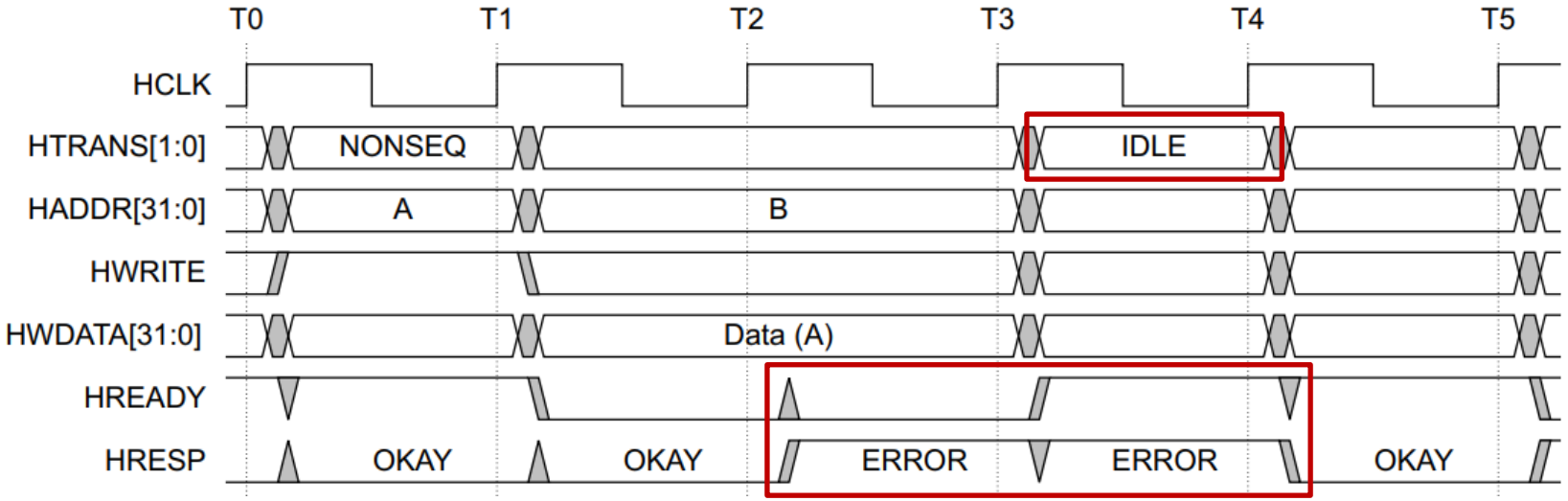
HRESP & HREADYOUT

	HRESP	HREADYOUT 0 Not Ready	HREADYOUT 1 Ready
OKAY	0	Transfer pending	Successful transfer completed
ERROR	1	ERROR response, first cycle	ERROR response, second cycle

- Pending
  - A subordinate uses HREADYOUT to insert a number of wait states
- Error
  - Attempt to write to a read-only memory location
  - If a NONSEQ or SEQ transfer is attempted to a nonexistent address location (OKAY response is given for IDLE and BUSY transfers)



Error Response



* Fixed pattern of error response

- ERROR: 2 cycles
- First error cycle: Extend the transfer for one extra cycle
- The two-cycle response provides sufficient time for the manager to cancel this next access and drive HTRANS[1:0] to IDLE before the start of the next transfer



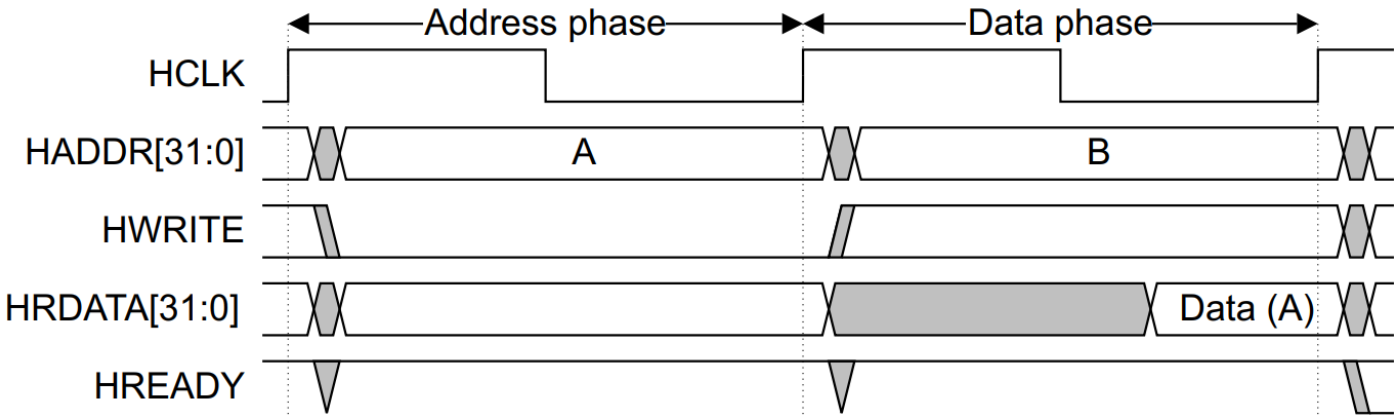
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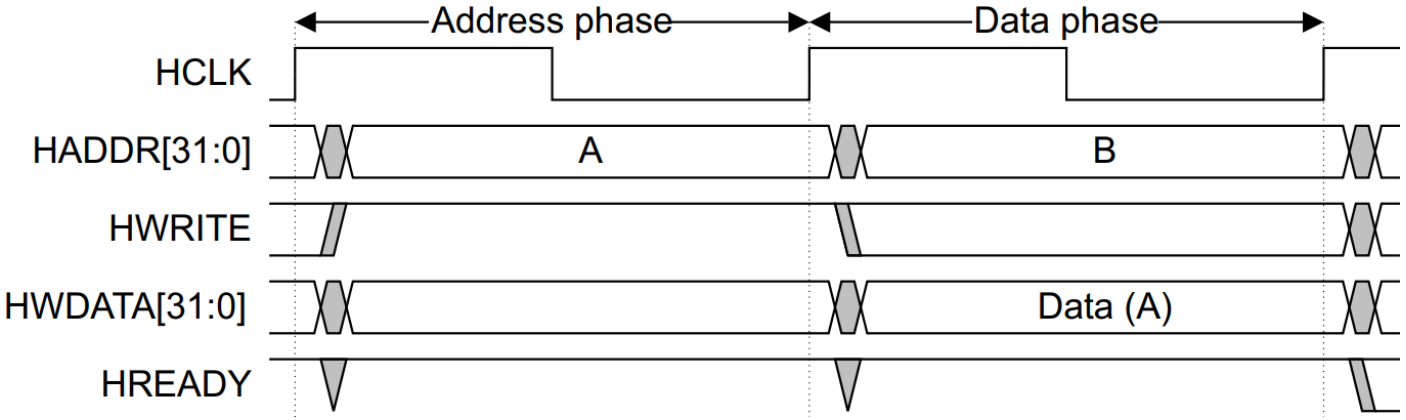


Basic Transfer: No Wait

Read Transfer



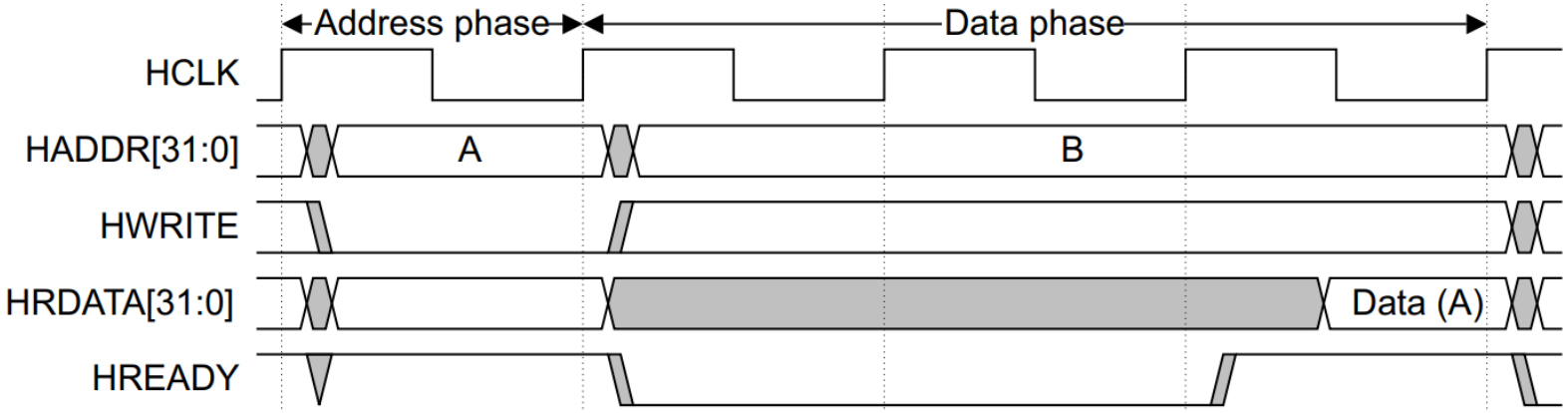
Write Transfer



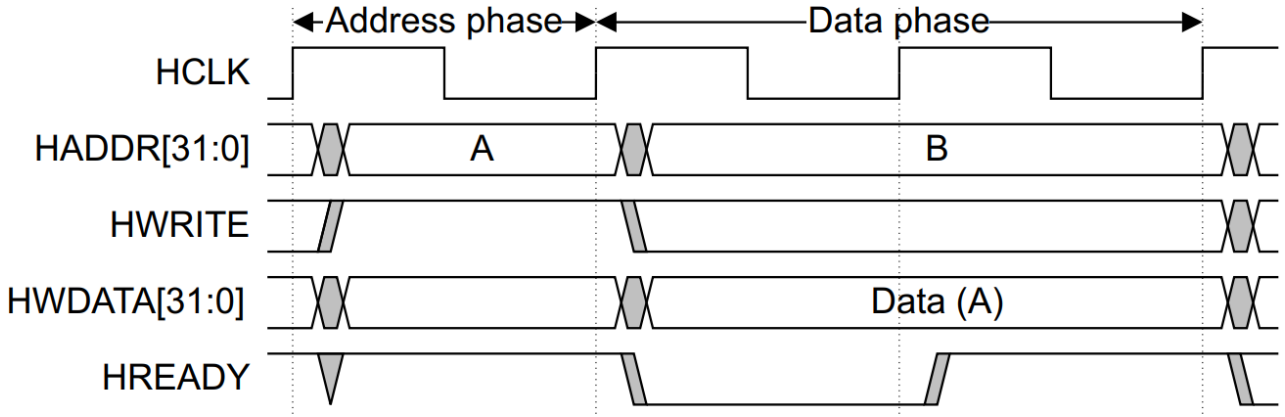


Transfer with Wait States

Read Transfer with Two Wait States

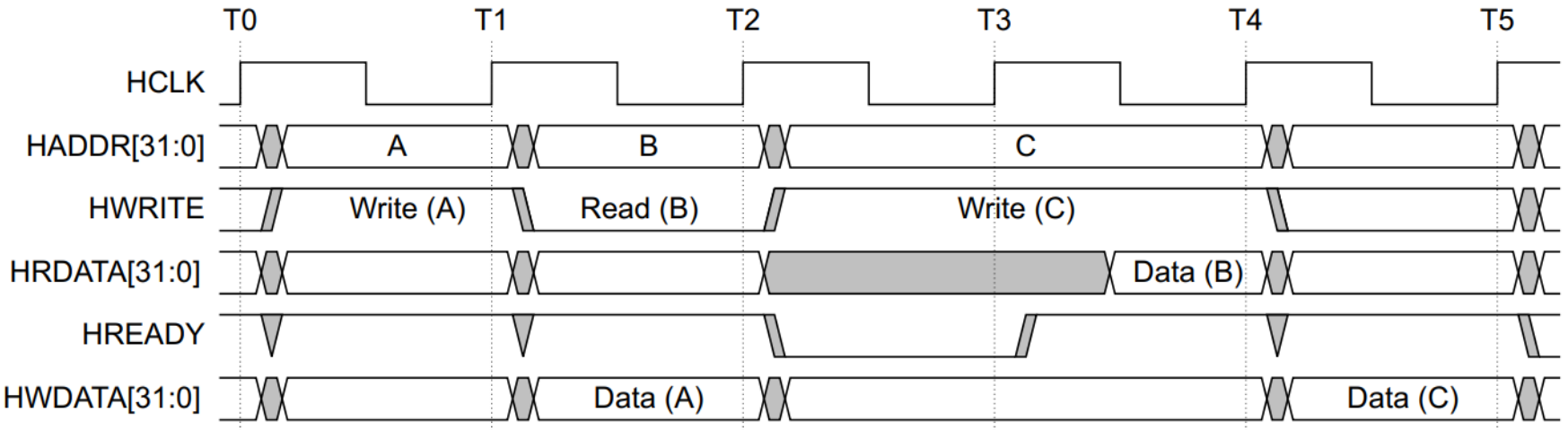


Write Transfer with One Wait State



Multiple Transfers

- The figure shows 3 transfers to unrelated addresses, A, B, and C with an extended address phase for address C



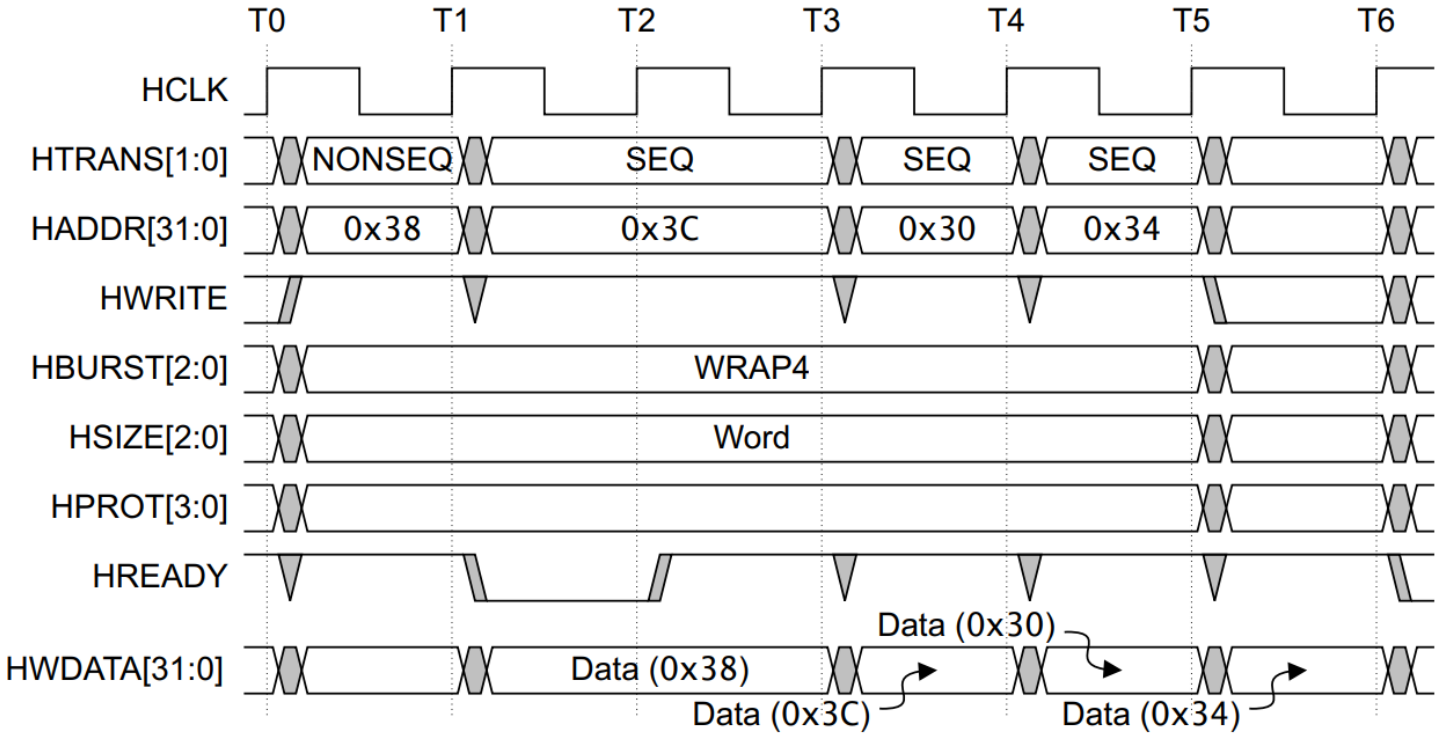


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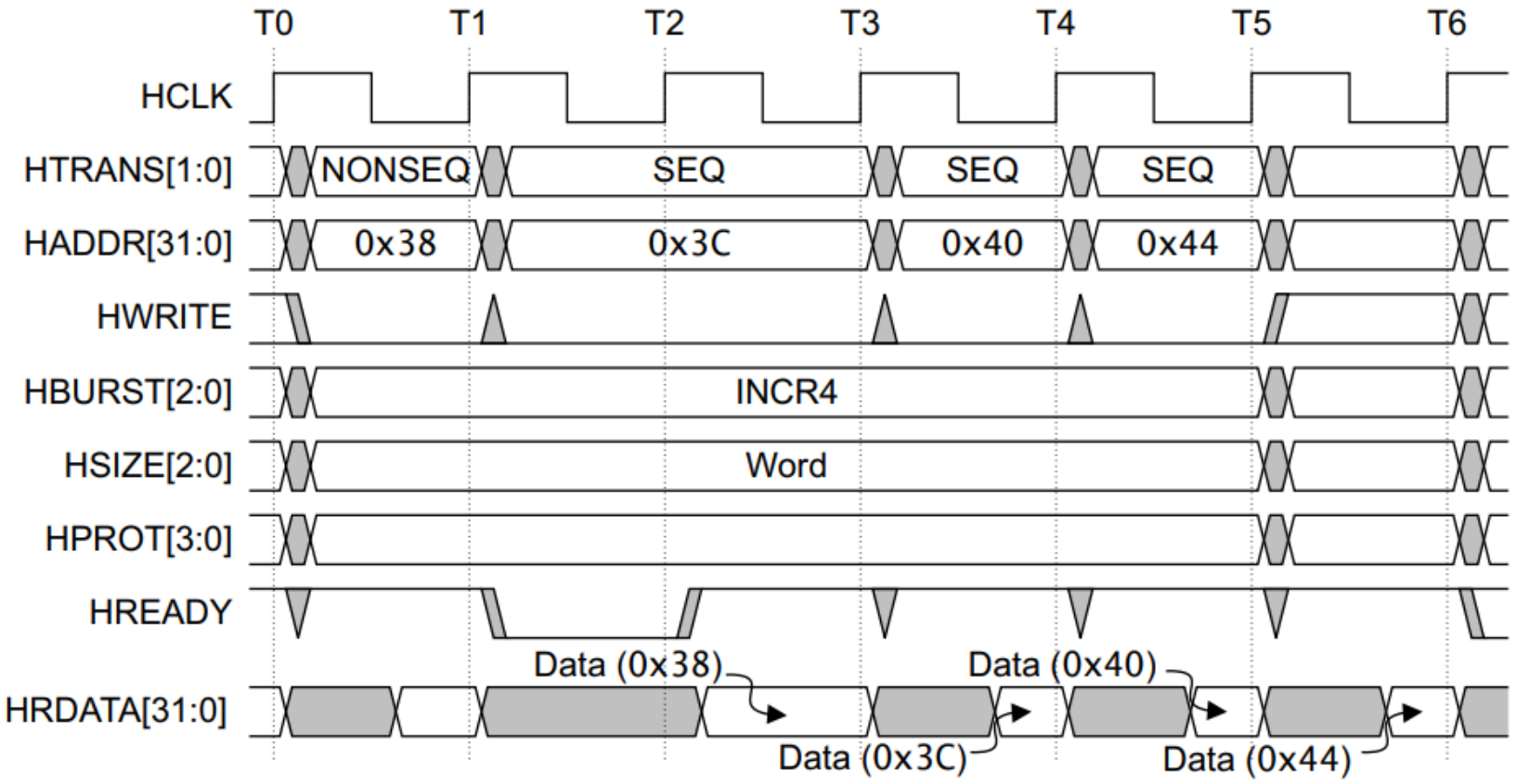


Burst Examples: WRAP4



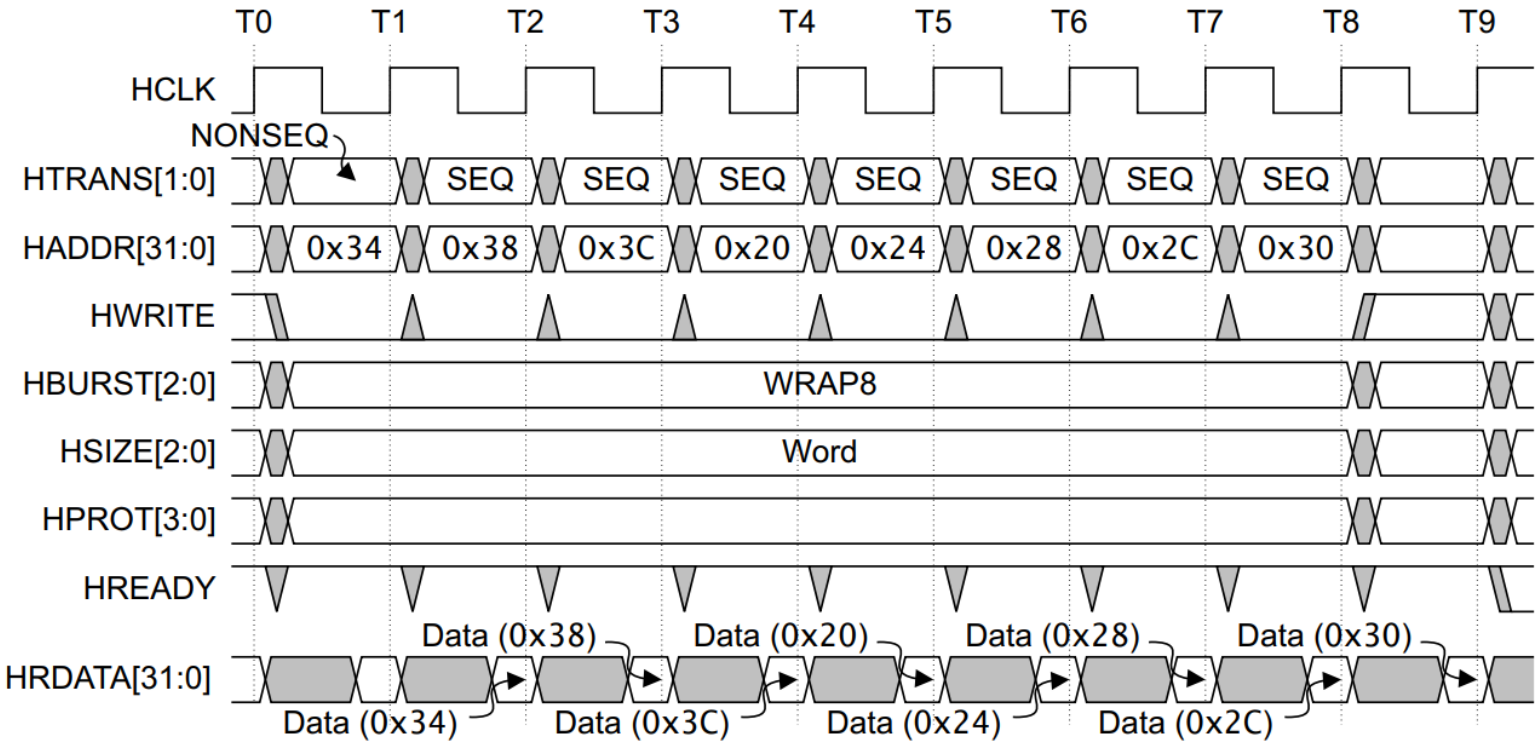
- Because the burst is a 4-beat burst of word transfers, the address wraps at 16-byte boundaries, and the transfer to address 0x3C is followed by a transfer to address 0x30

Burst Examples: INCR4





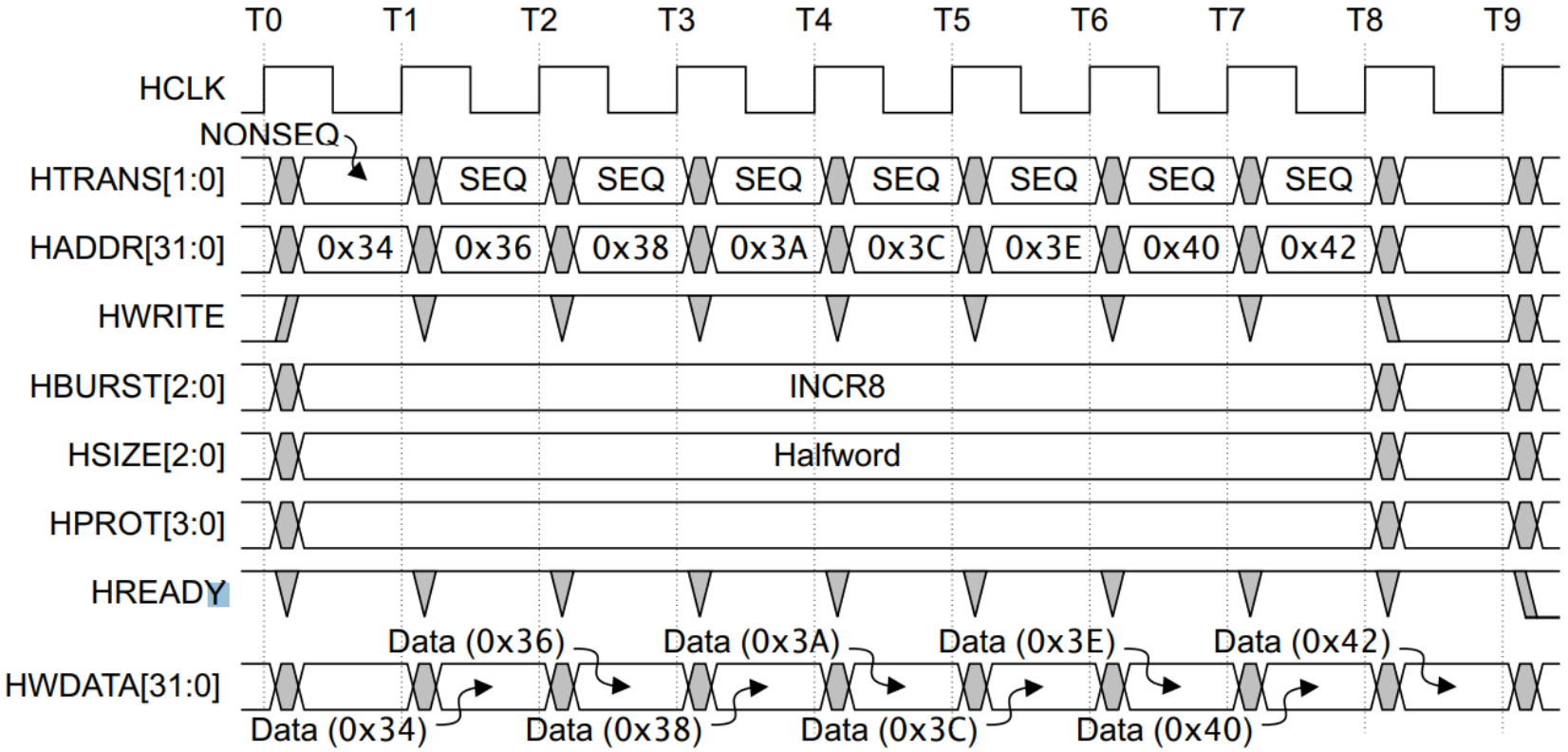
Burst Examples: WRAP8



- Since the burst is an 8-beat burst of word transfers, the address wraps at 32-byte boundaries, and the transfer to address 0x3C is followed by a transfer to address 0x20.



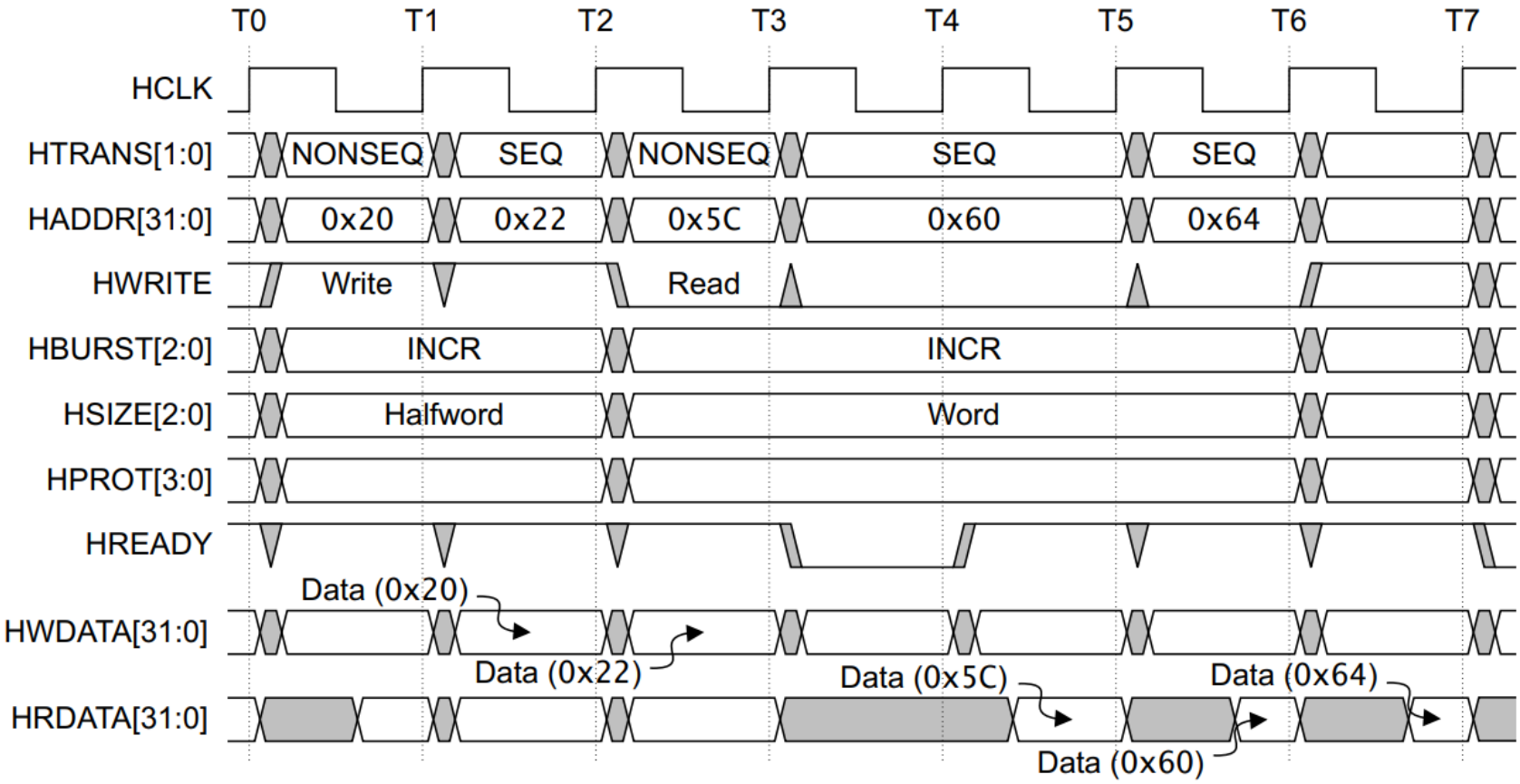
Burst Examples: INCR8



- The burst uses halfword transfers, therefore the addresses increase by two



Burst Examples: INCR (Undefined Length)





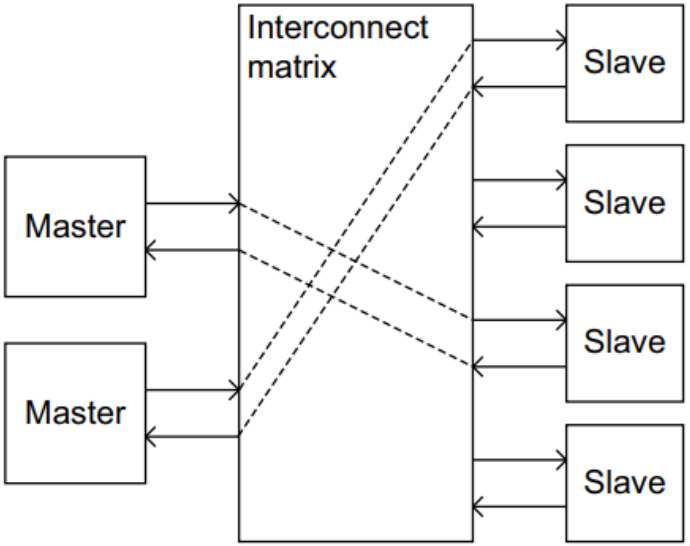
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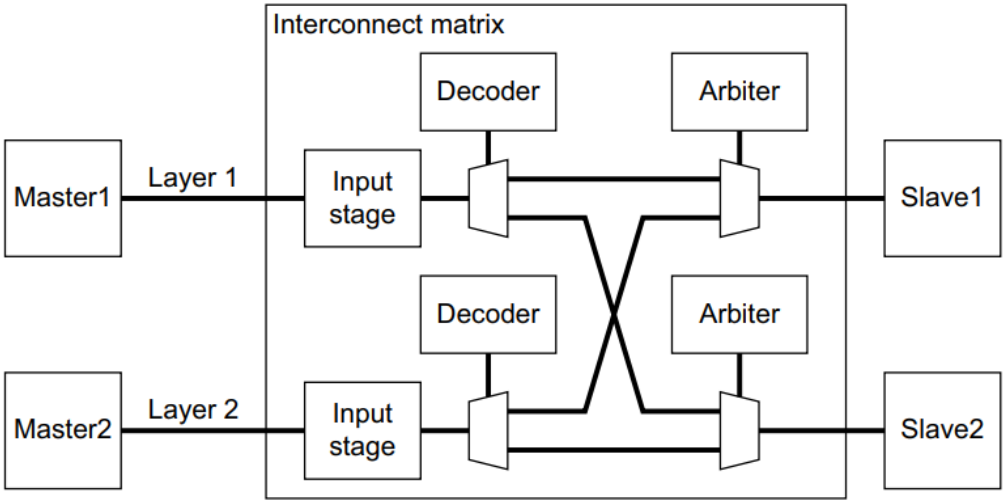
Multi-layer AHB

- A multi-layer AHB is an interconnection scheme that enables parallel access paths between multiple masters and slaves in a system
- Each AHB layer is simple because it only has one master, so no arbitration or master-to-slave multiplexing is required





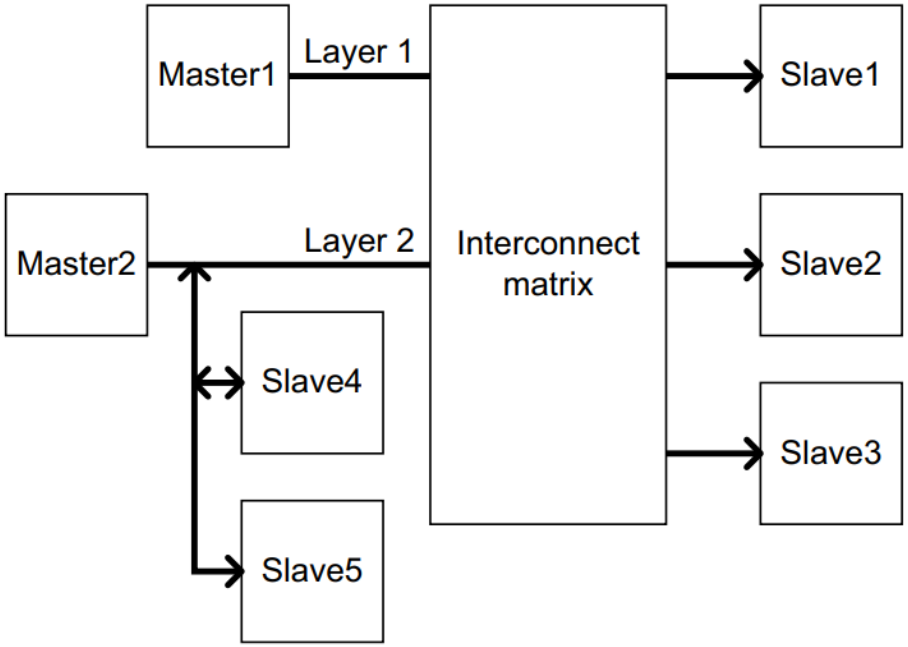
Implementation of Interconnect Matrix



- Every layer has a decode stage
- The layer that is not given access is waited using HREADY until it is given access to the required slave
- When a layer is waited, an input stage stores a copy of information until the access to the slave is given
- Each slave port has its own arbitration

Local Slaves

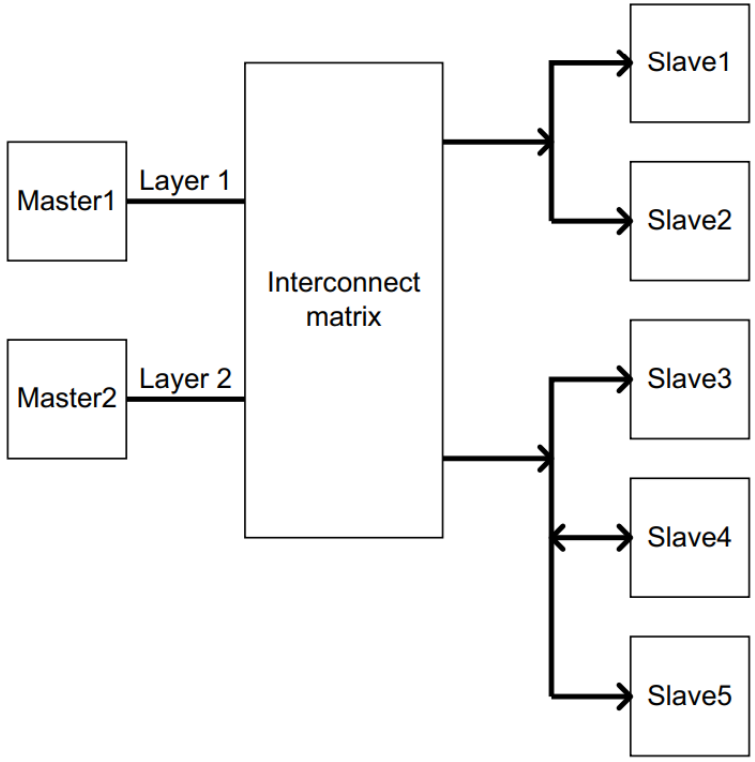
- The complexity of the interconnect matrix is reduced
- The slaves can be made local, or private, to a layer
- Scenario: A slave is only accessed by masters on the same layer





Multiple Slaves on One Slave Port

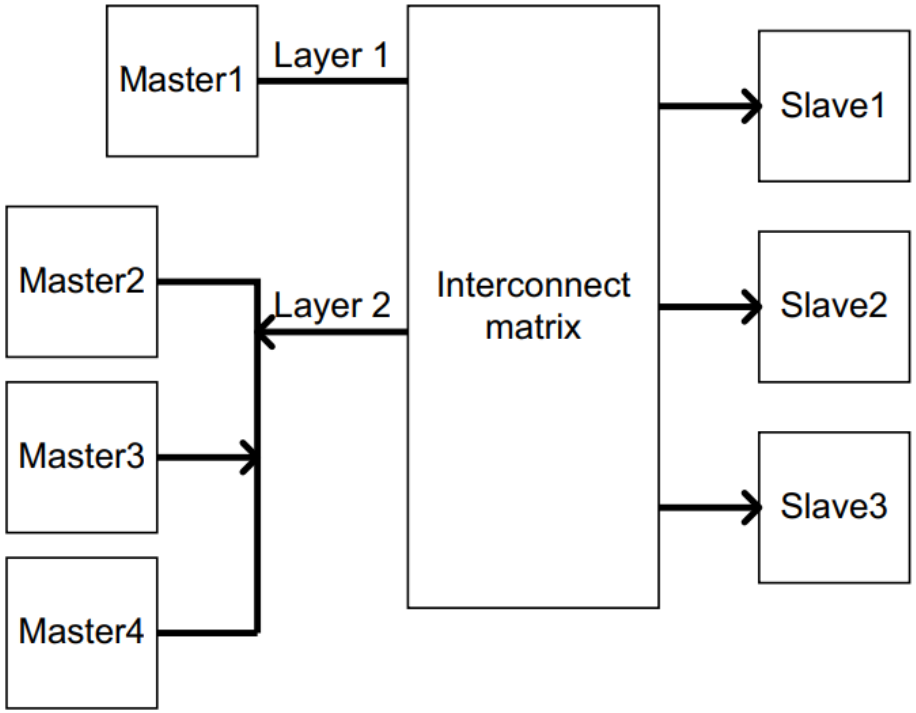
- It is useful to combine low-bandwidth slaves
- A set of slaves are normally accessed by just 1 master such as DMA controller





Multiple Masters on One Layer

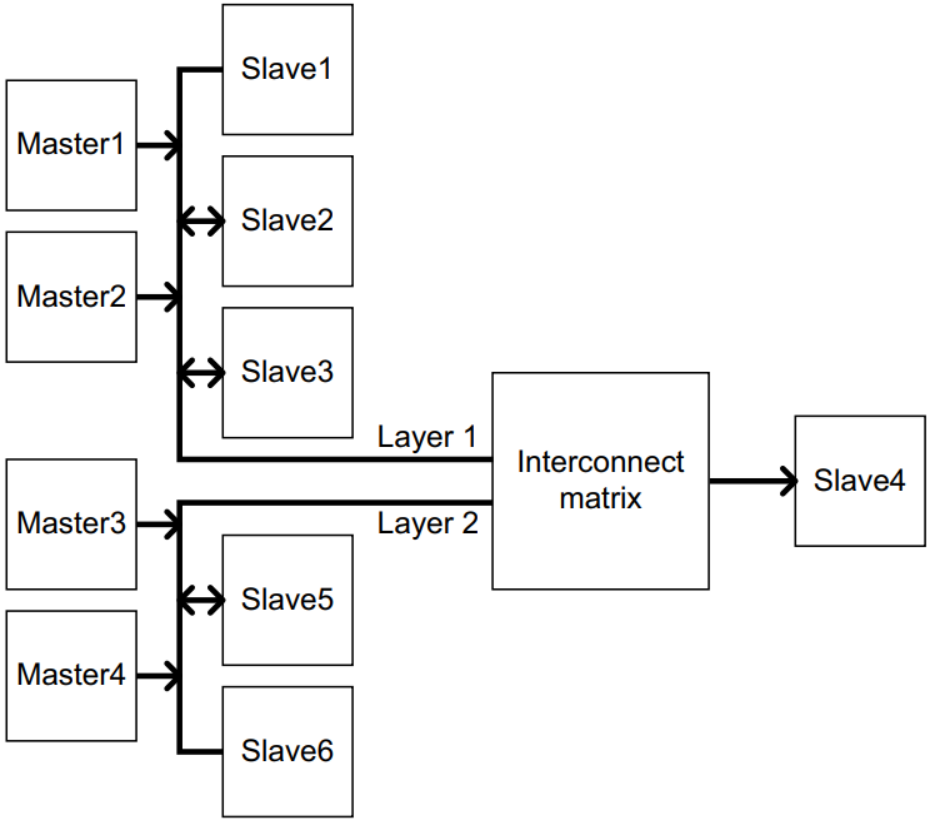
- Multiple masters share a layer
- This is well suited to combine low-bandwidth masters, such as the Test Interface Controller (TIC)





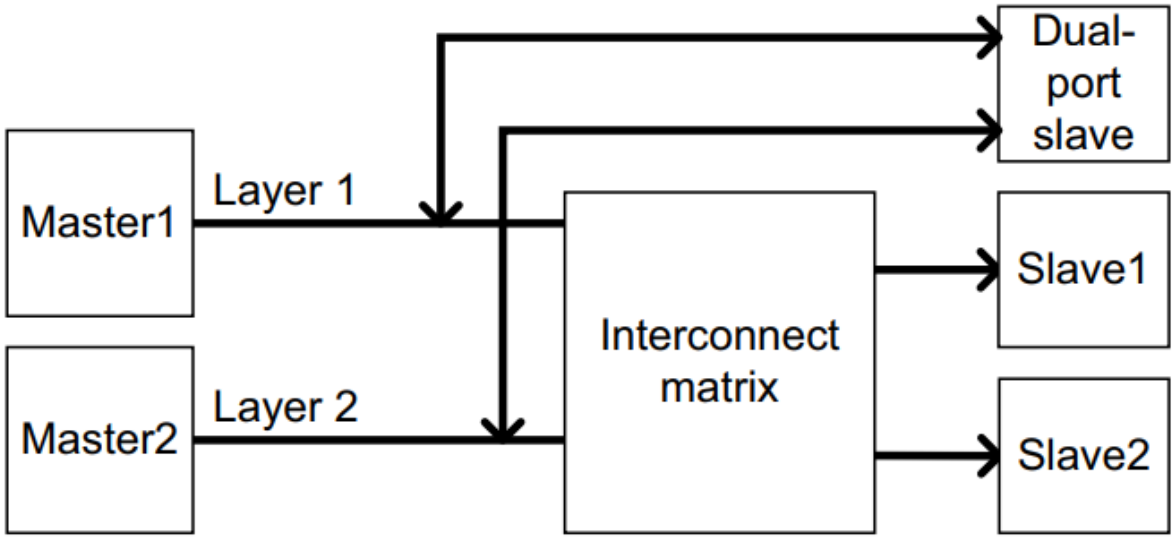
Separate AHB Subsystems

- Typically the shared Slave4 is an on-chip memory slave that is used as a buffer area between the two subsystems.



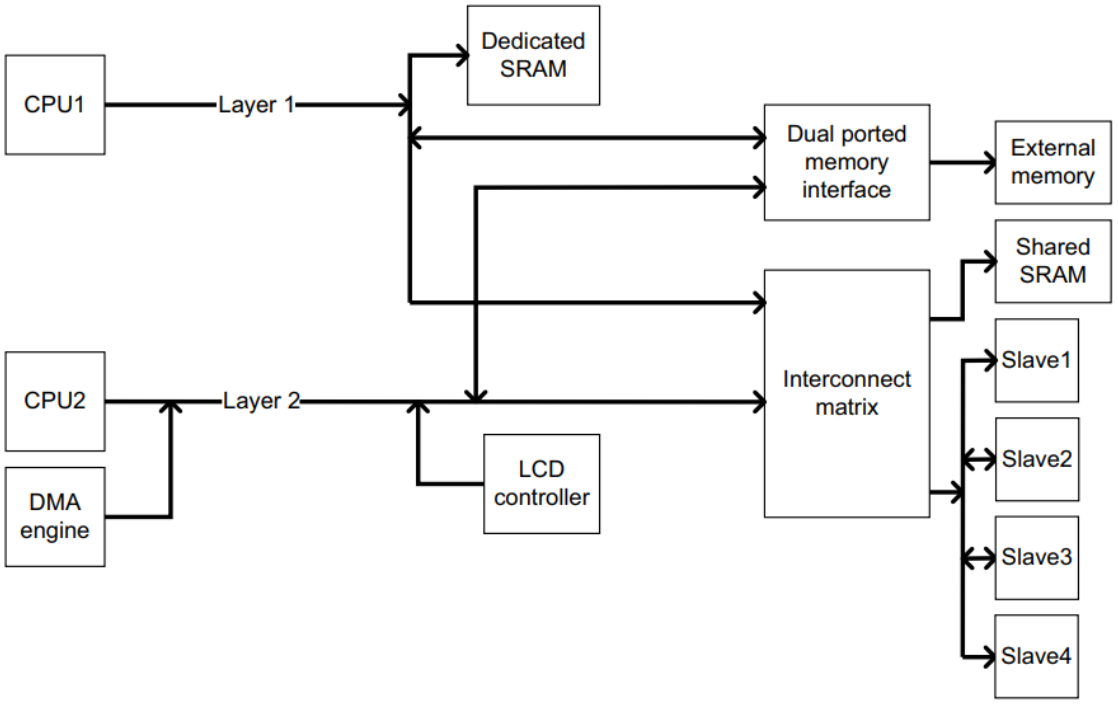
Multi-port Slaves

- The multi-port slaves, such as an SDRAM controller, are able to operate more efficiently by processing transfers from different layers in parallel





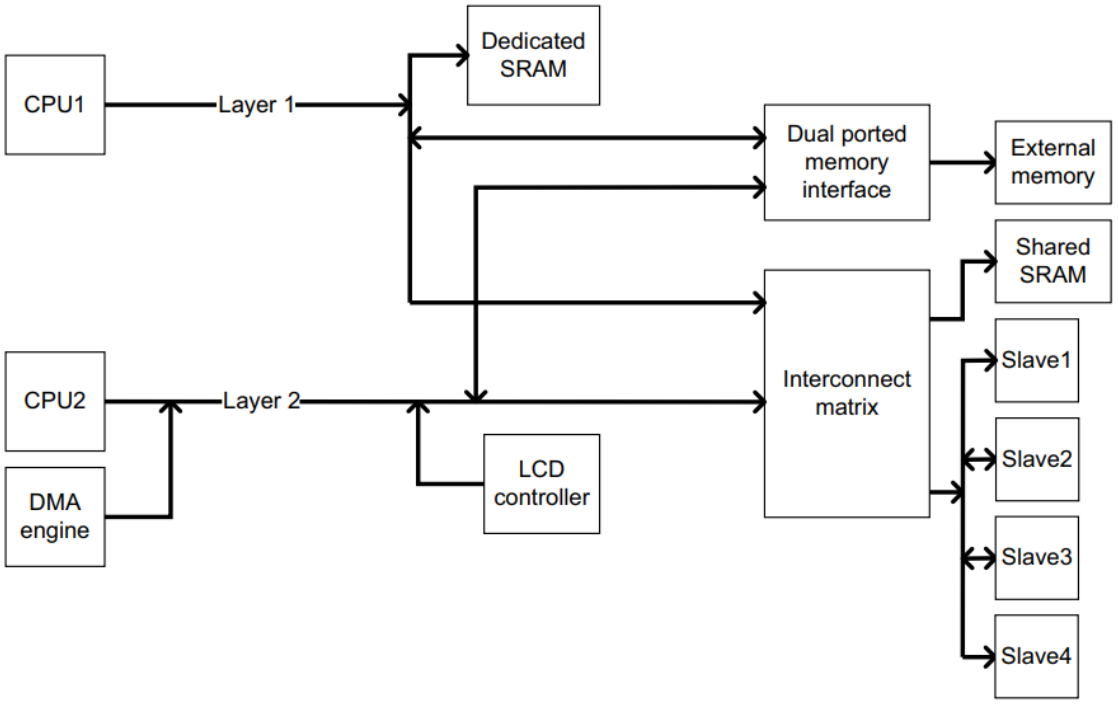
Example: Multi-layer AHB System (1/3)



- CPU1 has a dedicated AHB layer, layer 1
- CPU2 and the DMA engine share AHB layer 2
- The dedicated SRAM is only accessible from CPU1



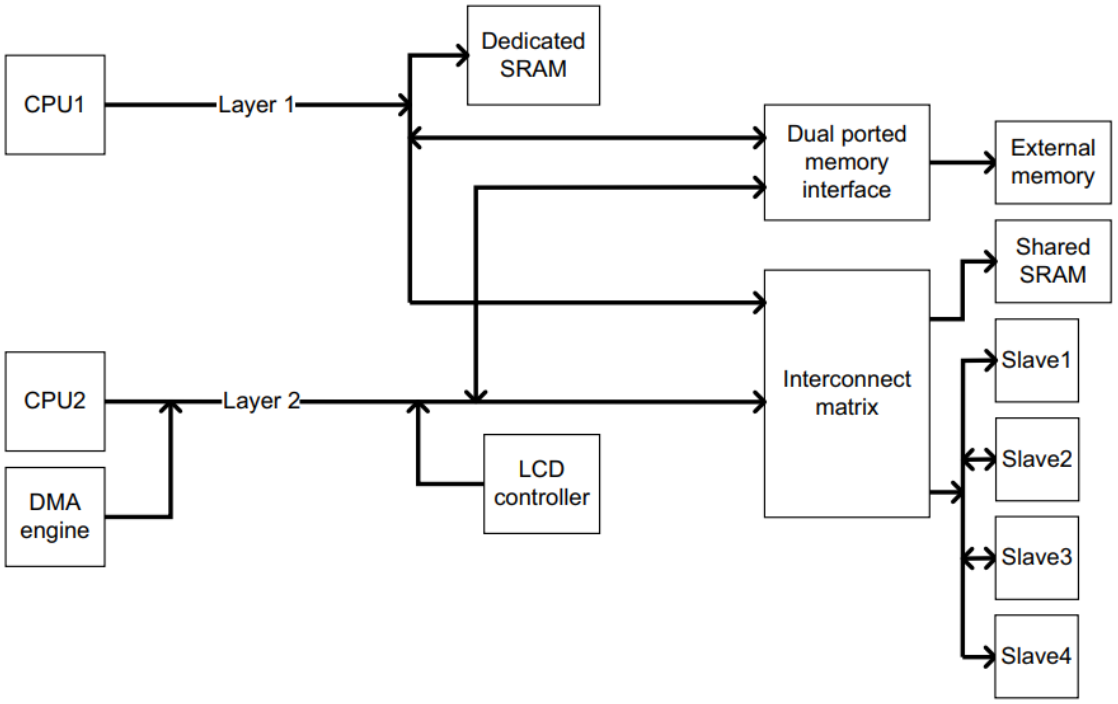
Example: Multi-layer AHB System (2/3)



- The LCD controller is only connected to layer 2, because this is the layer on which the DMA engine operates
- The external SDRAM interface has its own dual-port connection to both layers



Example: Multi-layer AHB System (3/3)



- The interconnect matrix has two slave ports, each accessible from either layer
- The slave ports can be an AHB to APB bridge, enabling the connection of many, low-bandwidth peripherals